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Practice Set 3

Your results are here!! for" Databricks Certified Data Engineer Associate Practice Test 3 "

0 of 50 questions answered correctly

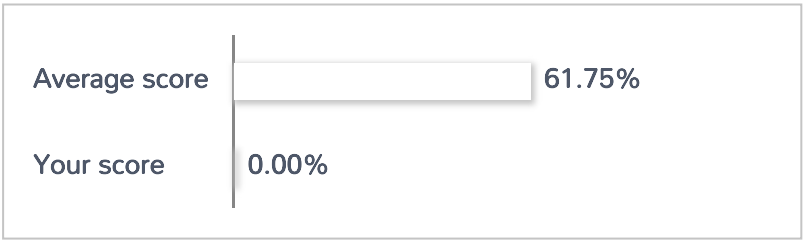
Your time: 00:00:23

Your Final Score is : 0

You have attempted : 0

Number of Correct Questions : 0 and scored 0

Number of Incorrect Questions : 0 and Negative marks 0



You can review your answers by clicking on “View Answers” option.
Important Note : Open Reference Documentation Links in New Tab (Right Click and Open in New Tab).

Restart Test View Answers

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
49 50

Answered Review

1. Question

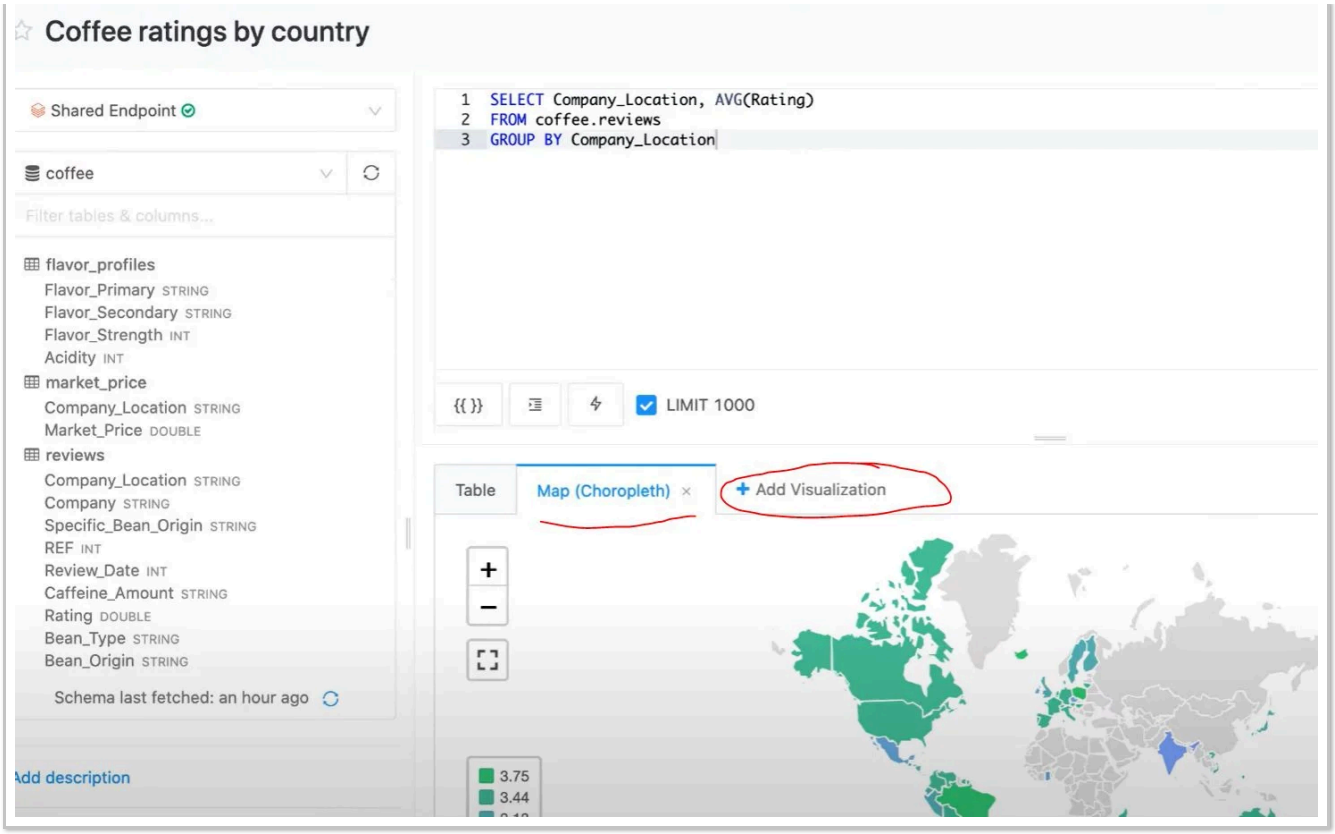
Which of the following is true, when building a Databricks SQL dashboard?

- A dashboard can only use results from one query
- Only one visualization can be developed with one query result
- A dashboard can only connect to one schema/Database
- More than one visualization can be developed using a single query result
- A dashboard can only have one refresh schedule

Unattempted

the answer is, More than one visualization can be developed using a single query result.

In the query editor pane + Add visualization tab can be used for many visualizations for a single query result.



2. Question

A newly joined team member John Smith in the Marketing team currently has access read access to sales tables but does not have access to update the table, which of the following commands help you accomplish this?

- GRANT UPDATE ON TABLE table_name TO john.smith@marketing.com
- GRANT USAGE ON TABLE table_name TO john.smith@marketing.com
- **GRANT MODIFY ON TABLE table_name TO john.smith@marketing.com**
- GRANT UPDATE TO TABLE table_name ON john.smith@marketing.com
- GRANT MODIFY TO TABLE table_name ON john.smith@marketing.com

Unattempted

The answer is GRANT MODIFY ON TABLE table_name TO john.smith@marketing.com

https://docs.microsoft.com/en-us/azure/databricks/security/access-control/table-acls/object-privileges#privileges

3. Question

A new user who currently does not have access to the catalog or schema is requesting access to the customer table in sales schema, but the customer table contains sensitive information, so you have decided to create view on the table excluding columns that are sensitive and granted access to the view GRANT SELECT ON view_name to user@company.com but when the user tries to query the view, gets the error view does not exist. What is the issue preventing user to access the view and how to fix it?

- User requires SELECT on the underlying table

- User requires to be put in a special group that has access to PII data
- User has to be the owner of the view
- **User requires USAGE privilege on Sales schema**
- User needs ADMIN privilege on the view

Unattempted

The answer is User requires USAGE privilege on Sales schema,

Data object privileges – Azure Databricks | Microsoft Docs

GRANT USAGE ON SCHEMA sales TO user@company.com;

USAGE: does not give any abilities, but is an additional requirement to perform any action on a schema object.

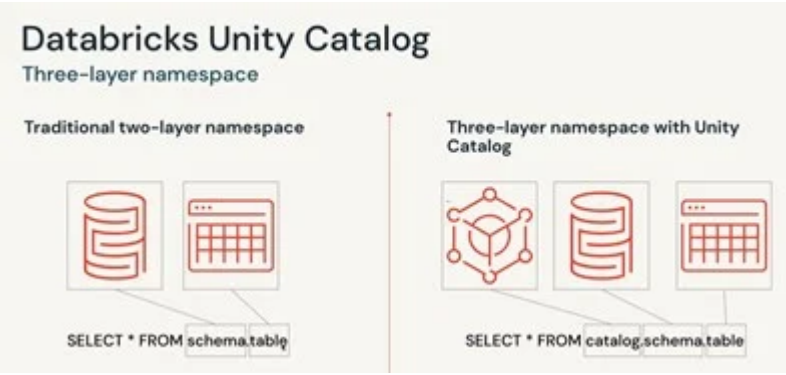
4. Question

How do you access or use tables in the unity catalog?

- schema_name.table_name
- schema_name.catalog_name.table_name
- catalog_name.table_name
- catalog_name.database_name.schema_name.table_name
- **catalog_name.schema_name.table_name**

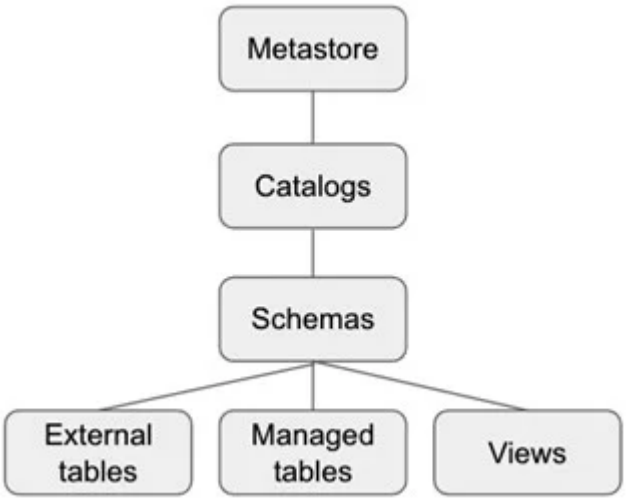
Unattempted

The answer is catalog_name.schema_name.table_name



note: Database and Schema are analogous they are interchangeably used in the Unity catalog.

FYI, A catalog is registered under a metastore, by default every workspace has a default metastore called hive_metastore, with a unity catalog you have the ability to create meatstores and share that across multiple workspaces.



5. Question

How do you upgrade an existing workspace managed table to a unity catalog table?
ALTER TABLE table_name SET UNITY_CATALOG = TRUE

- Create table catalog_name.schema_name.table_name
- as select * from hive_metastore.old_schema.old_table
- Create table table_name as select * from hive_metastore.old_schema.old_table
- Create table table_name format = UNITY as select * from old_table_name
- Create or replace table_name format = UNITY using deep clone old_table_name

Unattempted
The answer is Create table catalog_name.schema_name.table_name as select * from hive_metastore.old_schema.old_table
Basically, we are moving the data from an internal hive metastore to a metastore and catalog that is registered in the Unity catalog.
note: if it is a managed table the data is copied to a different storage account, for a large tables this can take a lot of time. For an external table the process is different.
Managed table: Upgrade a managed to Unity Catalog
External table: Upgrade an external table to Unity Catalog

6. Question

Which of the statements is correct when choosing between lakehouse and Datawarehouse?

- Traditional Data warehouses have special indexes which are optimized for Machine learning
- Traditional Data warehouses can serve low query latency with high reliability for BI workloads
- SQL support is only available for Traditional Datawarehouse's, Lakehouses support Python and Scala
- Traditional Data warehouses are the preferred choice if we need to support ACID, Lakehouse does not support ACID.

Lakehouse replaces the current dependency on data lakes and data warehouses uses an open standard storage format and supports low latency BI workloads.

Unattempted

The lakehouse replaces the current dependency on data lakes and data warehouses for modern data companies that desire:

· Open, direct access to data stored in standard data formats.

· Indexing protocols optimized for machine learning and data science.

· Low query latency and high reliability for BI and advanced analytics.

7. Question

Where are Interactive notebook results stored in Databricks product architecture?

Data plane

Control plane

Data and Control plane

JDBC data source

Databricks web application

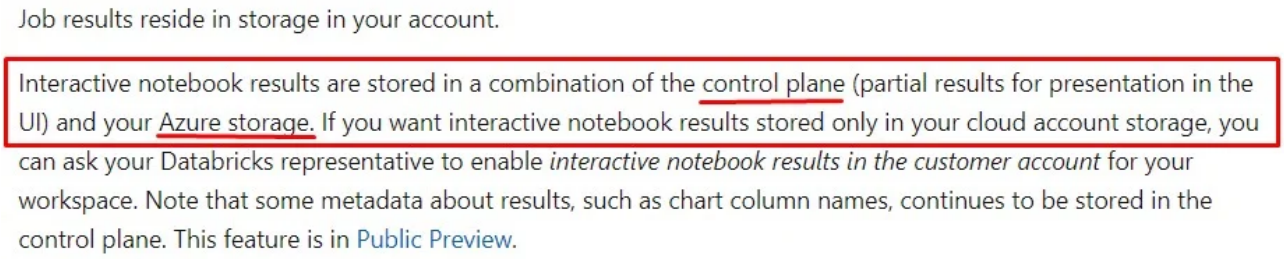
Unattempted

The answer is Data and Control plane,

Only Job results are stored in Data Plane(your storage), Interactive notebook results are stored in a combination of the control plane (partial results for presentation in the UI) and customer storage.

https://docs.microsoft.com/en-us/azure/databricks/getting-started/overview#—high-level-architecture

Snippet from the above documentation,

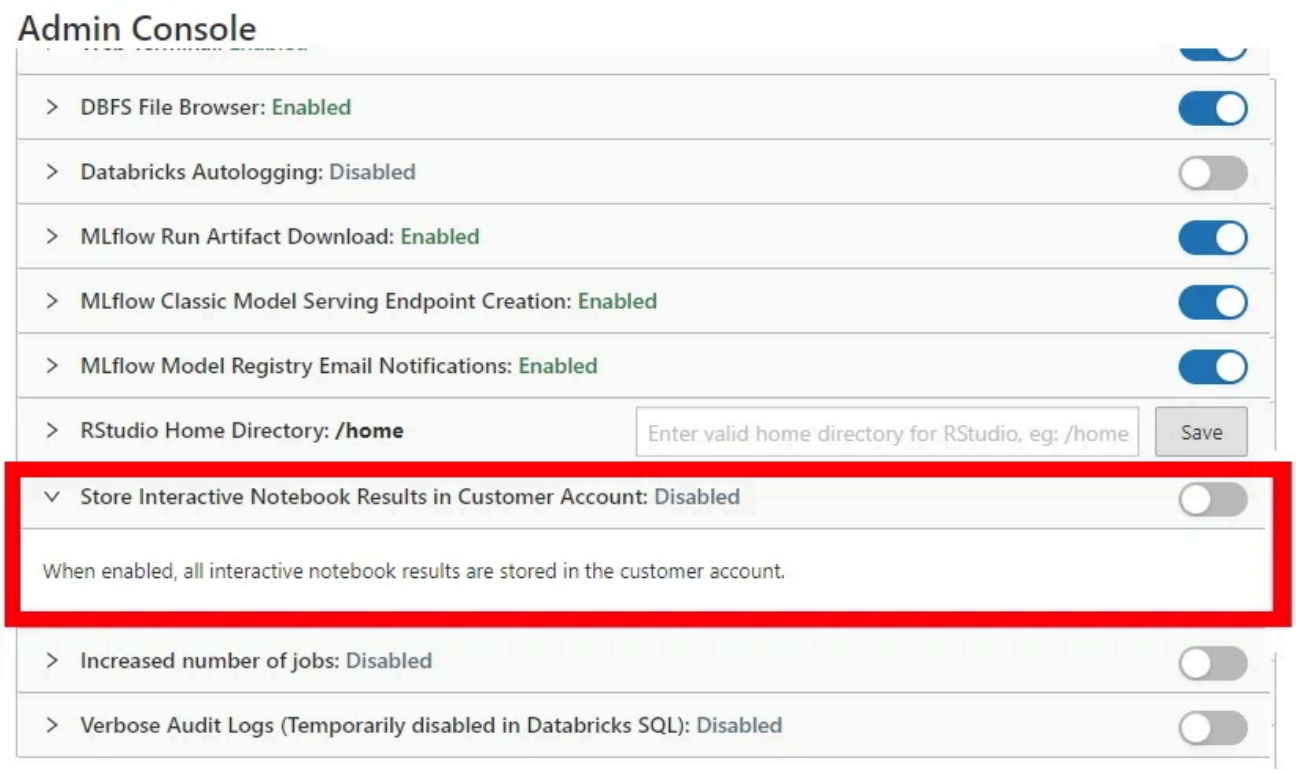


Job results reside in storage in your account.

Interactive notebook results are stored in a combination of the control plane (partial results for presentation in the UI) and your Azure storage. If you want interactive notebook results stored only in your cloud account storage, you can ask your Databricks representative to enable *interactive notebook results in the customer account* for your workspace. Note that some metadata about results, such as chart column names, continues to be stored in the control plane. This feature is in Public Preview.

How to change this behavior?

You can change this behavior using Workspace/Admin Console settings for that workspace, once enabled all of the interactive results are stored in the customer account(data plane) except the new notebook visualization feature Databricks has recently introduced, this still stores some metadata in the control pane irrespective of the below settings. please refer to the documentation for more details.



Why is this important to know?

I recently worked on a project where we had to deal with sensitive information of customers and we had a security requirement that all of the data need to be stored in the data plane including notebook results.

8. Question

Which of the following statements are true about a lakehouse?

- Lakehouse only supports Machine learning workloads and Data warehouses support BI workloads
- Lakehouse only supports end-to-end streaming workloads and Data warehouses support Batch workloads
- Lakehouse does not support ACID
- Lakehouse do not support SQL
- **Lakehouse supports Transactions**

Unattempted

What Is a Lakehouse? – The Databricks Blog

A lakehouse has the following key features:

- **Transaction support:** In an enterprise lakehouse many data pipelines will often be reading and writing data concurrently. Support for ACID transactions ensures consistency as multiple parties concurrently read or write data, typically using SQL.
- **Schema enforcement and governance:** The Lakehouse should have a way to support schema enforcement and evolution, supporting DW schema architectures such as star/snowflake-schemas. The system should be able to reason about data integrity, and it should have robust governance and auditing mechanisms.
- **BI support:** Lakehouses enable using BI tools directly on the source data. This reduces staleness and improves recency, reduces latency, and lowers the cost of having to operationalize two copies of the data in both a data lake and a warehouse.
- **Storage is decoupled from compute:** In practice this means storage and compute use separate clusters, thus these systems are able to scale to many more concurrent users and larger data sizes. Some modern data warehouses also have this property.
- **Openness:** The storage formats they use are open and standardized, such as Parquet, and they provide an API so a variety of tools and engines, including machine learning and Python/R libraries, can efficiently access the data **directly**.
- **Support for diverse data types ranging from unstructured to structured data**: The lakehouse can be used to store, refine, analyze, and access data types needed for many new data applications, including images, video, audio, semi-structured data, and text.
- **Support for diverse workloads:** including data science, machine learning, and SQL and analytics. Multiple tools might be needed to support all these workloads but they all rely on the same data repository.
- **End-to-end streaming:** Real-time reports are the norm in many enterprises. Support for streaming eliminates the need for separate systems dedicated to serving real-time data applications.

9. Question

Which of the following SQL command can be used to insert or update or delete rows based on a condition to check if a row(s) exists?

- MERGE INTO table_name
- COPY INTO table_name
- UPDATE table_name
- INSERT INTO OVERWRITE table_name
- INSERT IF EXISTS table_name

Unattempted

here is the additional documentation for your review.

https://docs.databricks.com/spark/latest/spark-sql/language-manual/delta-merge-into.html

MERGE INTO target_table_name [target_alias]

USING source_table_reference [source_alias]

ON merge_condition

[WHEN MATCHED [AND condition] THEN matched_action] [...]

[WHEN NOT MATCHED [AND condition] THEN not_matched_action] [...]

matched_action

{ DELETE |
UPDATE SET * |
UPDATE SET { column1 = value1 } [, ...] }
not_matched_action
{ INSERT * |
INSERT (column1 [, ...]) VALUES (value1 [, ...])

10. Question

When investigating a data issue you realized that a process accidentally updated the table, you want to query the same table with yesterday's version of the data so you can review what the prior version looks like, what is the best way to query historical data so you can do your analysis?

- SELECT * FROM TIME_TRAVEL(table_name) WHERE time_stamp = 'timestamp'
- TIME_TRAVEL FROM table_name WHERE time_stamp = date_sub(current_date(), 1)
- **SELECT * FROM table_name TIMESTAMP AS OF date_sub(current_date(), 1)**
- DISCRIBE HISTORY table_name AS OF date_sub(current_date(), 1)
- SHOW HISTORY table_name AS OF date_sub(current_date(), 1)

Unattempted
The answer is SELECT * FROM table_name TIMESTAMP as of date_sub(current_date(), 1)
FYI, Time travel supports two ways one is using timestamp and the second way is using version number,
Timestamp:
SELECT count(*) FROM my_table TIMESTAMP AS OF "2019-01-01"
SELECT count(*) FROM my_table TIMESTAMP AS OF date_sub(current_date(), 1)
SELECT count(*) FROM my_table TIMESTAMP AS OF "2019-01-01 01:30:00.000"
Version Number:
SELECT count(*) FROM my_table VERSION AS OF 5238
SELECT count(*) FROM my_table@v5238
SELECT count(*) FROM delta.'/path/to/my/table@v5238'
https://databricks.com/blog/2019/02/04/introducing-delta-time-travel-for-large-scale-data-lakes.html

11. Question

While investigating a data issue, you wanted to review yesterday's version of the table using below command, while querying the previous version of the table using time travel you realized that you are no longer able to view the historical data in the table and you could see it the table was updated yesterday based on the table history(DESCRIBE HISTORY table_name) command what could be the reason why you can not access this data?
SELECT * FROM table_name TIMESTAMP AS OF date_sub(current_date(), 1)

- You currently do not have access to view historical data
- By default, historical data is cleaned every 180 days in DELTA
- **A command VACUUM table_name RETAIN 0 was ran on the table**

- Time travel is disabled
- Time travel must be enabled before you query previous data

Unattempted

The answer is, VACUUM table_name RETAIN 0 was ran

The VACUUM command recursively vacuums directories associated with the Delta table and removes data files that are no longer in the latest state of the transaction log for the table and are older than a retention threshold. The default is 7 Days.

When VACUUM table_name RETAIN 0 is ran all of the historical versions of data are lost time travel can only provide the current state.

12. Question

You have accidentally deleted records from a table called transactions, what is the easiest way to restore the records deleted or the previous state of the table? Prior to deleting the version of the table is 3 and after delete the version of the table is 4.

- RESTORE TABLE transactions FROM VERSION as of 4
- **RESTORE TABLE transactions TO VERSION as of 3**
- INSERT INTO OVERWRITE transactions
SELECT * FROM transactions VERSION AS OF 3
- MINUS
- SELECT * FROM transactions
INSERT INTO OVERWRITE transactions
SELECT * FROM transactions VERSION AS OF 4
- INTERSECT

Unattempted

RESTORE (Databricks SQL) | Databricks on AWS

RESTORE [TABLE] table_name [TO] time_travel_version

Time travel supports using timestamp or version number

time_travel_version

{ TIMESTAMP AS OF timestamp_expression |

VERSION AS OF version }

timestamp_expression can be any one of:

'2018-10-18T22:15:12.013Z', that is, a string that can be cast to a timestamp

cast('2018-10-18 13:36:32 CEST' as timestamp)

'2018-10-18', that is, a date string

current_timestamp() – interval 12 hours

date_sub(current_date(), 1)

Any other expression that is or can be cast to a timestamp

13. Question

Create a schema called bronze using location '/mnt/delta/bronze', and check if the schema exists before creating.

- **CREATE SCHEMA IF NOT EXISTS bronze LOCATION '/mnt/delta/bronze'**
- CREATE SCHEMA bronze IF NOT EXISTS LOCATION '/mnt/delta/bronze'
- if IS_SCHEMA('bronze'): CREATE SCHEMA bronze LOCATION '/mnt/delta/bronze'
- Schema creation is not available in metastore, it can only be done in Unity catalog UI
- Cannot create schema without a database

Unattempted

https://docs.databricks.com/sql/language-manual/sql-ref-syntax-ddl-create-schema.html

CREATE SCHEMA [IF NOT EXISTS] schema_name [LOCATION schema_directory]

14. Question

How do you check the location of an existing schema in Delta Lake?

- Run SQL command SHOW LOCATION schema_name
- Check unity catalog UI
- Use Data explorer
- **Run SQL command DESCRIBE SCHEMA EXTENDED schema_name**
- Schemas are internally in-store external hive meta stores like MySQL or SQL Server

Unattempted

Here is an example of how it looks

```
%sql
DESCRIBE SCHEMA EXTENDED BRONZE
```

Table | Data Profile

	database_description_item	database_description_value
1	Namespace Name	BRONZE
2	Comment	Ths is a copy of raw data in DELTA format
3	Location	dbfs:/mnt/data/bronze
4	Owner	root
5	Properties	

Showing all 5 rows.

15. Question

Which of the below SQL commands create a Global temporary view?

- CREATE OR REPLACE TEMPORARY VIEW view_name
  AS SELECT * FROM table_name
- CREATE OR REPLACE LOCAL TEMPORARY VIEW view_name
  AS SELECT * FROM table_name
- **CREATE OR REPLACE GLOBAL TEMPORARY VIEW view_name**
  AS SELECT * FROM table_name
- CREATE OR REPLACE VIEW view_name
  AS SELECT * FROM table_name
- CREATE OR REPLACE LOCAL VIEW view_name
  AS SELECT * FROM table_name

Unattempted

CREATE OR REPLACE GLOBAL TEMPORARY VIEW view_name
AS SELECT * FROM table_name

There are two types of temporary views that can be created Local and Global

A session-scoped temporary view is only available with a spark session, so another notebook in the same cluster can not access it. if a notebook is detached and reattached local temporary view is lost.

A global temporary view is available to all the notebooks in the cluster but if a cluster restarts a global temporary view is lost.

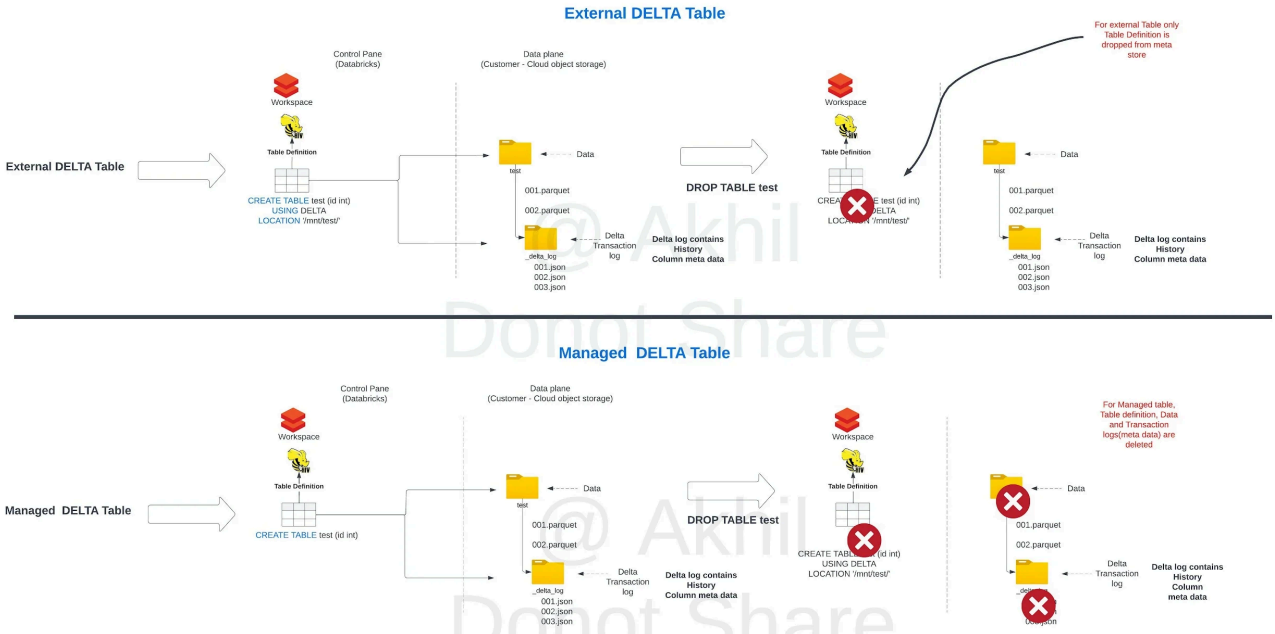
16. Question

When you drop a managed table using SQL syntax DROP TABLE table_name how does it impact metadata, history, and data stored in the table?

- Drops table from meta store, drops metadata, history, and data in storage.
- Drops table from meta store and data from storage but keeps metadata and history in storage
- Drops table from meta store, meta data and history but keeps the data in storage
- Drops table but keeps meta data, history and data in storage
- Drops table and history but keeps meta data and data in storage

Unattempted

For a managed table, a drop command will drop everything from metastore and storage.
See the below image to understand the differences between dropping an external table.



17. Question

The team has decided to take advantage of table properties to identify a business owner for each table, which of the following table DDL syntax allows you to populate a table property identifying the business owner of a table

- CREATE TABLE inventory (id INT, units FLOAT)
SET TBLPROPERTIES business_owner = 'supply chain'
- CREATE TABLE inventory (id INT, units FLOAT)
TBLPROPERTIES (business_owner = 'supply chain')
- CREATE TABLE inventory (id INT, units FLOAT)
SET (business_owner = 'supply chain')

CREATE TABLE inventory (id INT, units FLOAT)
SET PROPERTY (business_owner = 'supply chain')

CREATE TABLE inventory (id INT, units FLOAT)
SET TAG (business_owner = 'supply chain')

Unattempted
CREATE TABLE inventory (id INT, units FLOAT) TBLPROPERTIES (business_owner = 'supply chain')
Table properties and table options (Databricks SQL) | Databricks on AWS
Alter table command can used to update the TBLPROPERTIES
ALTER TABLE inventory SET TBLPROPERTIES(business_owner , 'operations')

18. Question

Data science team has requested they are missing a column in the table called average price, this can be calculated using units sold and sales amt, which of the following SQL statements allow you to reload the data with additional column

INSERT OVERWRITE sales
SELECT *, salesAmt/unitsSold as avgPrice FROM sales

CREATE OR REPLACE TABLE sales
AS SELECT *, salesAmt/unitsSold as avgPrice FROM sales

MERGE INTO sales USING (SELECT *, salesAmt/unitsSold as avgPrice FROM sales)

OVERWRITE sales AS SELECT *, salesAmt/unitsSold as avgPrice FROM sales

COPY INTO SALES AS SELECT *, salesAmt/unitsSold as avgPrice FROM sales

Unattempted
CREATE OR REPLACE TABLE sales
AS SELECT *, salesAmt/unitsSold as avgPrice FROM sales
The main difference between INSERT OVERWRITE and CREATE OR REPLACE TABLE(CRAS) is that CRAS can modify the schema of the table, i.e it can add new columns or change data types of existing columns. By default INSERT OVERWRITE only overwrites the data.
INSERT OVERWRITE can also be used to overwrite schema, only
when spark.databricks.delta.schema.autoMerge.enabled is set true if this option is not enabled and if there is a schema mismatch command will fail.

19. Question

You are working on a process to load external CSV files into a delta table by leveraging the COPY INTO command, but after running the command for the second time no data was loaded into the table name, why is that?
COPY INTO table_name
FROM 'dbfs:/mnt/raw/*.csv'
FILEFORMAT = CSV

- COPY INTO only works one time data load
- Run REFRESH TABLE sales before running COPY INTO
- **COPY INTO did not detect new files after the last load**
- Use incremental = TRUE option to load new files
- COPY INTO does not support incremental load, use AUTO LOADER

Unattempted

The answer is COPY INTO did not detect new files after the last load,
COPY INTO keeps track of files that were successfully loaded into the table, the next time when the COPY INTO runs it skips them.
FYI, you can change this behavior by using COPY_OPTIONS 'force'= 'true', when this option is enabled all files in the path/pattern are loaded.
COPY INTO table_identifier
FROM [file_location | (SELECT identifier_list FROM file_location)]
FILEFORMAT = data_source
[FILES = [file_name, ... | PATTERN = 'regex_pattern']
[FORMAT_OPTIONS ('data_source_reader_option' = 'value', ...)]
[COPY_OPTIONS 'force' = ('false'|'true')]

20. Question

What is the main difference between the below two commands?
INSERT OVERWRITE table_name
SELECT * FROM table
CREATE OR REPLACE TABLE table_name
AS SELECT * FROM table

- **INSERT OVERWRITE replaces data by default, CREATE OR REPLACE replaces data and Schema by default**
- INSERT OVERWRITE replaces data and schema by default, CREATE OR REPLACEreplaces data by default
- INSERT OVERWRITE maintains historical data versions by default, CREATE OR REPLACEclears the historical data versions by default
- INSERT OVERWRITE clears historical data versions by default, CREATE OR REPLACE maintains the historical data versions by default
- Both are same and results in identical outcomes

Unattempted

The answer is, INSERT OVERWRITE replaces data, CRAS replaces data and Schema
The main difference between INSERT OVERWRITE and CREATE OR REPLACE TABLE(CRAS) is that CRAS can modify the schema of the table, i.e it can add new columns or change data types of existing columns. By default INSERT OVERWRITE only overwrites the data.

INSERT OVERWRITE can also be used to overwrite schema, only
when spark.databricks.delta.schema.autoMerge.enabled is set true if this option is not enabled and if there is a schema mismatch command will fail.

21. Question

Which of the following functions can be used to convert JSON string to Struct data type?

- TO_STRUCT (json value)
- FROM_JSON (json value)
- **FROM_JSON (json value, schema of json)**
- CONVERT (json value, schema of json)
- CAST (json value as STRUCT)

Unattempted
Syntax
Copy
from_json(jsonStr, schema [, options])
Arguments
jsonStr: A STRING expression specifying a row of CSV data.
schema: A STRING literal or invocation of schema_of_json function (Databricks SQL).
options: An optional MAP literal specifying directives.
Refer documentation for more details,
https://docs.microsoft.com/en-us/azure/databricks/sql/language-manual/functions/from_json

22. Question

You are working on a marketing team request to identify customers with the same information between two tables CUSTOMERS_2021 and CUSTOMERS_2020 each table contains 25 columns with the same schema, You are looking to identify rows that match between two tables across all columns, which of the following can be used to perform in SQL

- SELECT * FROM CUSTOMERS_2021
UNION
SELECT * FROM CUSTOMERS_2020
- SELECT * FROM CUSTOMERS_2021
UNION ALL
SELECT * FROM CUSTOMERS_2020
- SELECT * FROM CUSTOMERS_2021 C1
INNER JOIN CUSTOMERS_2020 C2
ON C1.CUSTOMER_ID = C2.CUSTOMER_ID
- **SELECT * FROM CUSTOMERS_2021**
INTERSECT

SELECT * FROM CUSTOMERS_2020

SELECT * FROM CUSTOMERS_2021
EXCEPT
SELECT * FROM CUSTOMERS_2020

Unattempted

Answer is,

SELECT * FROM CUSTOMERS_2021
INTERSECT
SELECT * FROM CUSTOMERS_2020

To compare all the rows between both the tables across all the columns using intersect will help us achieve that, an inner join is only going to check if the same column value exists across both the tables on a single column.

INTERSECT [ALL | DISTINCT]

Returns the set of rows which are in both subqueries.

If ALL is specified a row that appears multiple times in the subquery1 as well as in subquery will be returned multiple times.

If DISTINCT is specified the result does not contain duplicate rows. This is the default.

23. Question

You are looking to process the data based on two variables, one to check if the department is supply chain and second to check if process flag is set to True

if department = “supply chain” & process:

if department == “supply chain” && process:

if department == “supply chain” & process == TRUE:

if department == “supply chain” & if process == TRUE:

if department == “supply chain“ and process:

Unattempted

24. Question

You were asked to create a notebook that can take department as a parameter and process the data accordingly, which is the following statements result in storing the notebook parameter into a python variable

SET department = dbutils.widget.get(“department“)

ASSIGN department == dbutils.widget.get(“department“)

department = dbutils.widget.get(“department“)

department = notebook.widget.get(“department“)

department = notebook.param.get(“department“)

Unattempted

The answer is department = dbutils.widget.get(“department“)

Refer to additional documentation here

https://docs.databricks.com/notebooks/widgets.html

25. Question

Which of the following statements can successfully read the notebook widget and pass the python variable to a SQL statement in a Python notebook cell?

order_date = dbutils.widgets.get(“widget_order_date“)
spark.sql(f“SELECT * FROM sales WHERE orderDate = ‘f{order_date }‘“)

order_date = dbutils.widgets.get(“widget_order_date“)
spark.sql(f“SELECT * FROM sales WHERE orderDate = ‘order_date‘ “)

order_date = dbutils.widgets.get(“widget_order_date“)
spark.sql(f”SELECT * FROM sales WHERE orderDate = ‘${order_date }‘ “)

order_date = dbutils.widgets.get(“widget_order_date“)
spark.sql(f“SELECT * FROM sales WHERE orderDate = ‘{order_date}‘ “)

order_date = dbutils.widgets.get(“widget_order_date“)
spark.sql(“SELECT * FROM sales WHERE orderDate = order_date“)

Unattempted

26. Question

The below spark command is looking to create a summary table based customerId and the number of times the customerId is present in the event_log delta table and write a one-time micro-batch to a summary table, fill in the blanks to complete the query.

spark._____
.format(“delta“)
.table(“events_log“)
.groupBy(“customerId“)
.count()
._____
.format(“delta“)
.outputMode(“complete“)
.option(“checkpointLocation“, “/tmp/delta/eventsByCustomer/_checkpoints/“)
.trigger(_____)
.table(“target_table“)

writeStream, readStream, once

readStream, writeStream, once

writeStream, processingTime = once

writeStream, readStream, once = True

readStream, writeStream, once = True

Unattempted

The answer is readStream, writeStream, once = True.

```
spark.readStream
  .format("delta")
  .table("events_log")
  .groupBy("customerId")
  .count()
  .writeStream
  .format("delta")
  .outputMode("complete")
  .option("checkpointLocation", "/tmp/delta/eventsByCustomer/_checkpoints/")
  .trigger(once = True)
  .table("target_table")
```

27. Question

You would like to build a spark streaming process to read from a Kafka queue and write to a Delta table every 15 minutes, what is the correct trigger option

trigger("15 minutes")

trigger(process "15 minutes")

trigger(processingTime = 15)

trigger(processingTime = "15 Minutes")

trigger(15)

Unattempted

The answer is trigger(processingTime = "15 Minutes")

Triggers:

Unspecified

This is the default. This is equivalent to using processingTime="500ms"

Fixed interval micro-batches .trigger(processingTime="2 minutes")

The query will be executed in micro-batches and kicked off at the user-specified intervals

One-time micro-batch .trigger(once=True)

The query will execute a single micro-batch to process all the available data and then stop on its own

One-time micro-batch.trigger .trigger(availableNow=True) —— New feature a better version of (once=True)

Databricks supports trigger(availableNow=True) in Databricks Runtime 10.2 and above for Delta Lake and Auto Loader sources. This functionality combines the batch processing approach of trigger once with the ability to configure batch size, resulting in multiple parallelized batches that give greater control for right-sizing batches and the resultant files.

28. Question

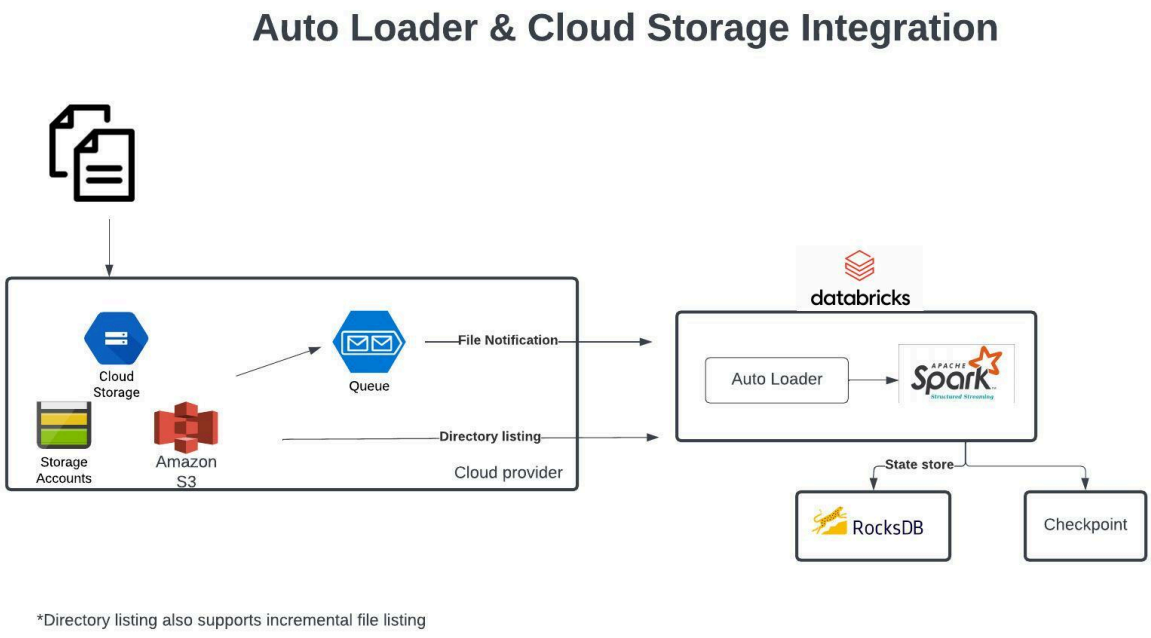
Which of the following scenarios is the best fit for the AUTO LOADER solution?

- Efficiently process new data incrementally from cloud object storage
- Incrementally process new streaming data from Apache Kafa into delta lake
- Incrementally process new data from relational databases like MySQL
- Efficiently copy data from data lake location to another data lake location
- Efficiently move data incrementally from one delta table to another delta table

Unattempted

The answer is, Efficiently process new data incrementally from cloud object storage.

Please note: AUTO LOADER only works on data/files located in cloud object storage like S3 or Azure Blob Storage it does not have the ability to read other data sources, although AUTO LOADER is built on top of structured streaming it only supports files in the cloud object storage. If you want to use Apache Kafka then you can just use structured streaming.



Auto Loader and Cloud Storage Integration

Auto Loader supports a couple of ways to ingest data incrementally

1. Directory listing – List Directory and maintain the state in RocksDB, supports incremental file listing

2. File notification – Uses a trigger+queue to store the file notification which can be later used to retrieve the file, unlike Directory listing File notification can scale up to millions of files per day.

[OPTIONAL]

Auto Loader vs COPY INTO?

Auto Loader

Auto Loader incrementally and efficiently processes new data files as they arrive in cloud storage without any additional setup. Auto Loader provides a new Structured Streaming source called cloudFiles. Given an input directory path on the cloud file storage, the cloudFiles source automatically processes new files as they arrive, with the option of also processing existing files in that directory.

When to use Auto Loader instead of the COPY INTO?

You want to load data from a file location that contains files in the order of millions or higher. Auto Loader can discover files more efficiently than the COPY INTO SQL command and can split file processing into multiple batches.

You do not plan to load subsets of previously uploaded files. With Auto Loader, it can be more difficult to reprocess subsets of files. However, you can use the COPY INTO SQL command to reload subsets of files while an Auto Loader stream is simultaneously running.

Refer to more documentation here,

https://docs.microsoft.com/en-us/azure/databricks/ingestion/auto-loader

29. Question

You had AUTO LOADER to process millions of files a day and noticed slowness in load process, so you scaled up the Databricks cluster but realized the performance of the Auto loader is still not improving, what is the best way to resolve this.

- AUTO LOADER is not suitable to process millions of files a day
- Setup a second AUTO LOADER process to process the data
- **Increase the maxFilesPerTrigger option to a sufficiently high number**
- Copy the data from cloud storage to local disk on the cluster for faster access
- Merge files to one large file

Unattempted

The default value of maxFilesPerTrigger is 1000 it can be increased to a much higher number but will require a much larger compute to process.

cloudFiles.maxFilesPerTrigger

Type: `Integer`

The maximum number of new files to be processed in every trigger. When used together with `cloudFiles.maxBytesPerTrigger`, Databricks consumes up to the lower limit of `cloudFiles.maxFilesPerTrigger` or `cloudFiles.maxBytesPerTrigger`, whichever is reached first. This option has no effect when used with `Trigger.Once()`.

Default value: 1000

https://docs.databricks.com/ingestion/auto-loader/options.html

30. Question

The current ELT pipeline is receiving data from the operations team once a day so you had setup an AUTO LOADER process to run once a day using trigger (Once = True) and scheduled a job to run once a day,

operations team recently rolled out a new feature that allows them to send data every 1 min, what changes do you need to make to AUTO LOADER to process the data every 1 min.

- Convert AUTO LOADER to structured streaming
- **Change AUTO LOADER trigger to .trigger(ProcessingTime = "1 minute")**
- Setup a job cluster run the notebook once a minute
- Enable stream processing
- Change AUTO LOADER trigger to ("1 minute")

Unattempted

31. Question

What is the purpose of the bronze layer in a Multi-hop Medallion architecture?

- **Copy of raw data, easy to query and ingest data for downstream processes.**
- Powers ML applications
- Data quality checks, corrupt data quarantined
- Contain aggregated data that is to be consumed into Silver
- Reduces data storage by compressing the data

Unattempted

The answer is, copy of raw data, easy to query and ingest data for downstream processes,

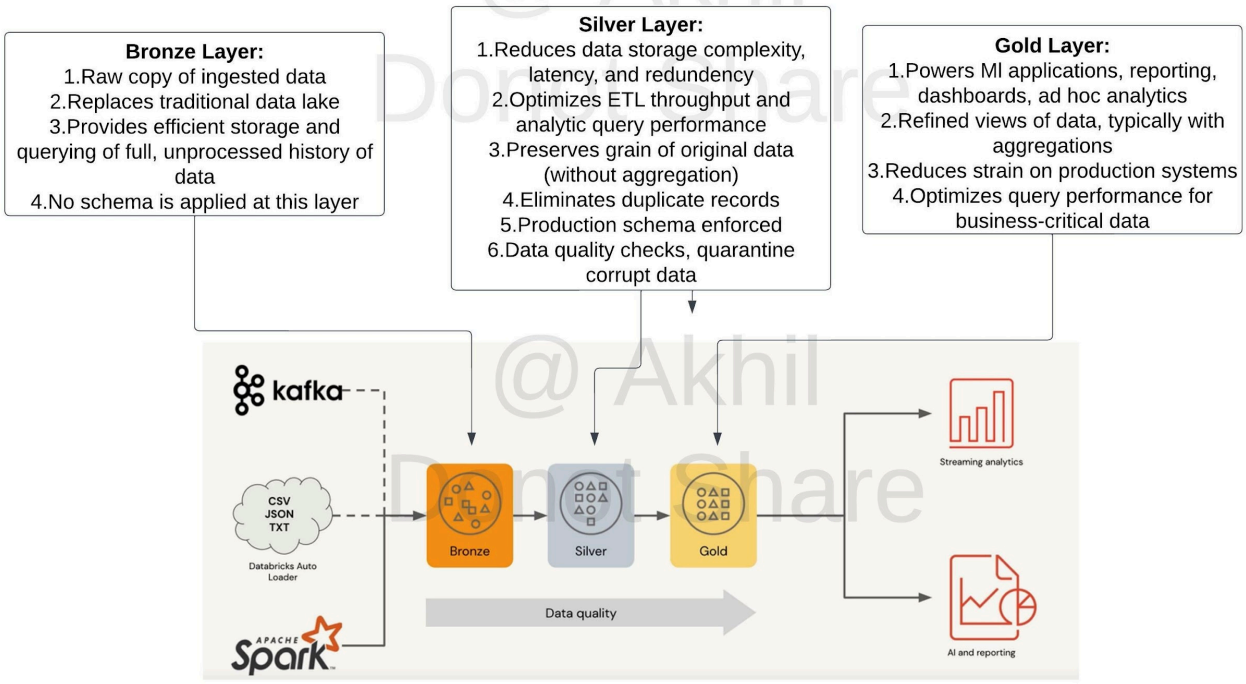
Medallion Architecture – Databricks

Here are the typical role of Bronze Layer in a medallion architecture.

Bronze Layer:

1. Raw copy of ingested data
2. Replaces traditional data lake
3. Provides efficient storage and querying of full, unprocessed history of data
4. No schema is applied at this layer

Exam focus: Please review the below image and understand the role of each layer(bronze, silver, gold) in medallion architecture, you will see varying questions targeting each layer and its purpose.



32. Question

What is the purpose of the silver layer in a Multi hop architecture?

- Replaces a traditional data lake
- Efficient storage and querying of full, unprocessed history of data
- **Eliminates duplicate data, quarantines bad data**
- Refined views with aggregated data
- Optimized query performance for business-critical data

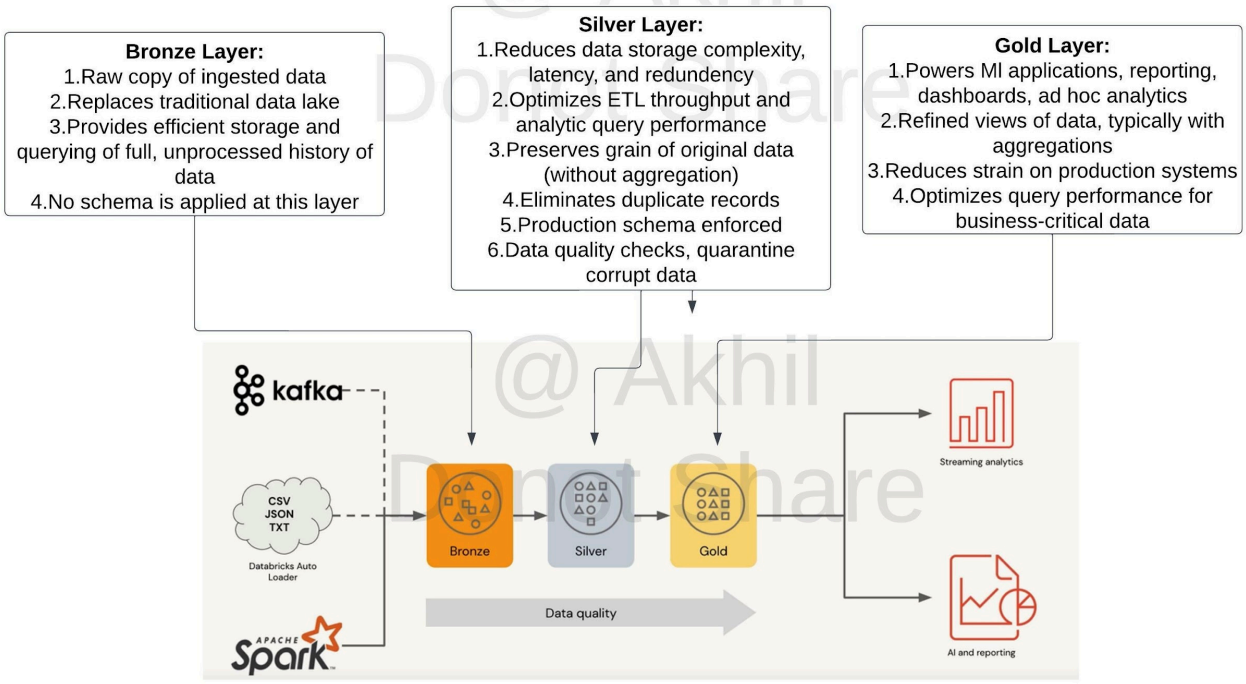
Unattempted

Medallion Architecture – Databricks

Silver Layer:

1. Reduces data storage complexity, latency, and redundency
2. Optimizes ETL throughput and analytic query performance
3. Preserves grain of original data (without aggregation)
4. Eliminates duplicate records
5. production schema enforced
6. Data quality checks, quarantine corrupt data

Exam focus: Please review the below image and understand the role of each layer(bronze, silver, gold) in medallion architecture, you will see varying questions targeting each layer and its purpose.



33. Question

What is the purpose of gold layer in Multi hop architecture?

- Optimizes ETL throughput and analytic query performance
- Eliminate duplicate records
- Preserves grain of original data, without any aggregations
- Data quality checks and schema enforcement
- **Optimized query performance for business-critical data**

Unattempted

Medallion Architecture – Databricks

Gold Layer:

1. Powers MI applications, reporting, dashboards, ad hoc analytics
2. Refined views of data, typically with aggregations
3. Reduces strain on production systems
4. Optimizes query performance for business-critical data

Exam focus: Please review the below image and understand the role of each layer(bronze, silver, gold) in medallion architecture, you will see varying questions targeting each layer and its purpose.

34. Question

The Delta Live Tables Pipeline is configured to run in Development mode using the Triggered Pipeline Mode. what is the expected outcome after clicking Start to update the pipeline?

All datasets will be updated once and the pipeline will shut down. The compute resources will be terminated

All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will be deployed for the update and terminated when the pipeline is stopped

All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will persist after the pipeline is stopped to allow for additional development and testing

All datasets will be updated once and the pipeline will shut down. The compute resources will persist to allow for additional development and testing

All datasets will be updated continuously and the pipeline will not shut down. The compute resources will persist with the pipeline

Unattempted

The answer is All datasets will be updated once and the pipeline will shut down. The compute resources will persist to allow for additional testing.

DLT pipeline supports two modes Development and Production, you can switch between the two based on the stage of your development and deployment lifecycle.

Development and production modes

When you run your pipeline in development mode, the Delta Live Tables system:

Reuses a cluster to avoid the overhead of restarts.

Disables pipeline retries so you can immediately detect and fix errors.

In production mode, the Delta Live Tables system:

Restarts the cluster for specific recoverable errors, including memory leaks and stale credentials.

Retries execution in the event of specific errors, for example, a failure to start a cluster.

Use the 

buttons in the Pipelines UI to switch between development and production modes. By default, pipelines run in development mode.

Switching between development and production modes only controls cluster and pipeline execution behavior. Storage locations must be configured as part of pipeline settings and are not affected when switching between modes.

Please review additional DLT concepts using below link

https://docs.databricks.com/data-engineering/delta-live-tables/delta-live-tables-concepts.html#delta-live-tables-concepts

35. Question

The Delta Live Table Pipeline is configured to run in Production mode using the continuous Pipeline Mode. what is the expected outcome after clicking Start to update the pipeline?

All datasets will be updated once and the pipeline will shut down. The compute resources will be terminated

All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will be deployed for the update and terminated when the pipeline is stopped

All datasets will be updated at set intervals until the pipeline is shut down. The compute resources will persist after the pipeline is stopped to allow for additional testing

All datasets will be updated once and the pipeline will shut down. The compute resources will persist to allow for additional testing

All datasets will be updated continuously and the pipeline will not shut down. The compute resources will persist with the pipeline

Unattempted

The answer is,

All datasets will be updated continuously and the pipeline will not shut down. The compute resources will persist with the pipeline until it is shut down since the execution mode is chosen to be continuous. It does not matter if the pipeline mode is development or production, pipeline mode only matters during the pipeline initialization.

DLT pipeline supports two modes Development and Production, you can switch between the two based on the stage of your development and deployment lifecycle.

Development and production modes

Development:

When you run your pipeline in development mode, the Delta Live Tables system:

Reuses a cluster to avoid the overhead of restarts.

Disables pipeline retries so you can immediately detect and fix errors.

Production:

In production mode, the Delta Live Tables system:

Restarts the cluster for specific recoverable errors, including memory leaks and stale credentials.

Retries execution in the event of specific errors, for example, a failure to start a cluster.

Use the 

buttons in the Pipelines UI to switch between development and production modes. By default, pipelines run in development mode.

Switching between development and production modes only controls cluster and pipeline execution behavior. Storage locations must be configured as part of pipeline settings and are not affected when switching between modes.

Delta Live Tables supports two different modes of execution:

Triggered pipelines update each table with whatever data is currently available and then stop the cluster running the pipeline. Delta Live Tables automatically analyzes the dependencies between your tables and starts by computing those that read from external sources. Tables within the pipeline are updated after their dependent data sources have been updated.

Continuous pipelines update tables continuously as input data changes. Once an update is started, it continues to run until manually stopped. Continuous pipelines require an always-running cluster but ensure that downstream consumers have the most up-to-date data

Please review additional DLT concepts using the below link

https://docs.databricks.com/data-engineering/delta-live-tables/delta-live-tables-concepts.html#delta-live-tables-concepts

36. Question

You are working to set up two notebooks to run on a schedule, the second notebook is dependent on the first notebook but both notebooks need different types of compute to run in an optimal fashion, what is the best way to set up these notebooks as jobs?

- Use DELTA LIVE PIPELINES instead of notebook tasks
- A Job can only use single cluster, setup job for each notebook and use job dependency to link both jobs together
- **Each task can use different cluster, add these two notebooks as two tasks in a single job with linear dependency and modify the cluster as needed for each of the tasks**
- Use a single job to setup both notebooks as individual tasks, but use the cluster API to setup the second cluster before the start of second task
- Use a very large cluster to run both the tasks in a single job

Unattempted

Tasks in Jobs support different clusters for each task in the same job.

37. Question

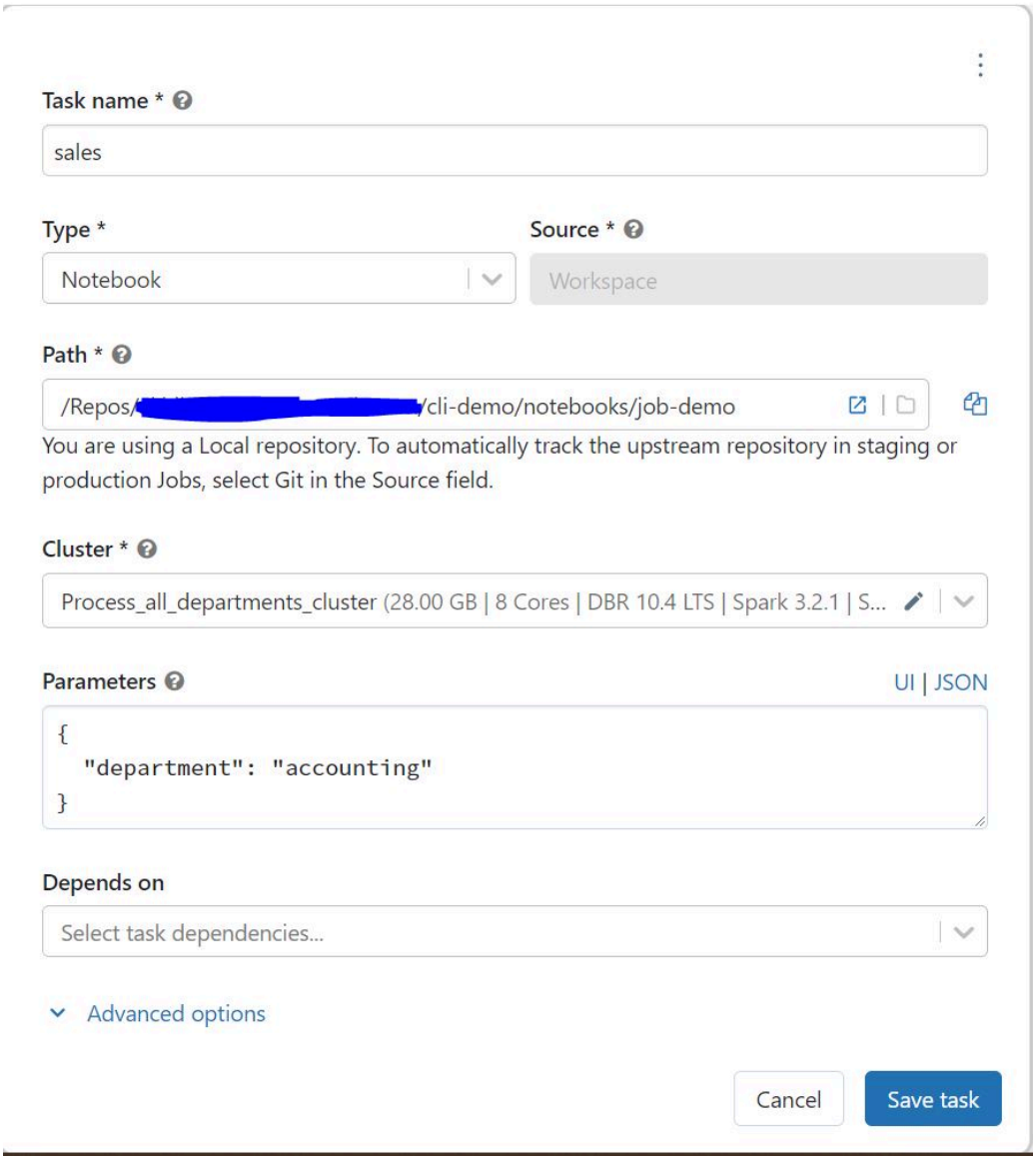
You are tasked to set up a set notebook as a job for six departments and each department can run the task parallelly, the notebook takes an input parameter dept number to process the data by department, how do you go about to setup this up in job?

- Use a single notebook as task in the job and use dbutils.notebook.run to run each notebook with parameter in a different cell
- A task in the job cannot take an input parameter, create six notebooks with hardcoded dept number and setup six tasks with linear dependency in the job
- **A task accepts key-value pair parameters, creates six tasks pass department number as parameter foreach task with no dependency in the job as they can all run in parallel.**
- A parameter can only be passed at the job level, create six jobs pass department number to each job with linear job dependency
- A parameter can only be passed at the job level, create six jobs pass department number to each job with no job dependency

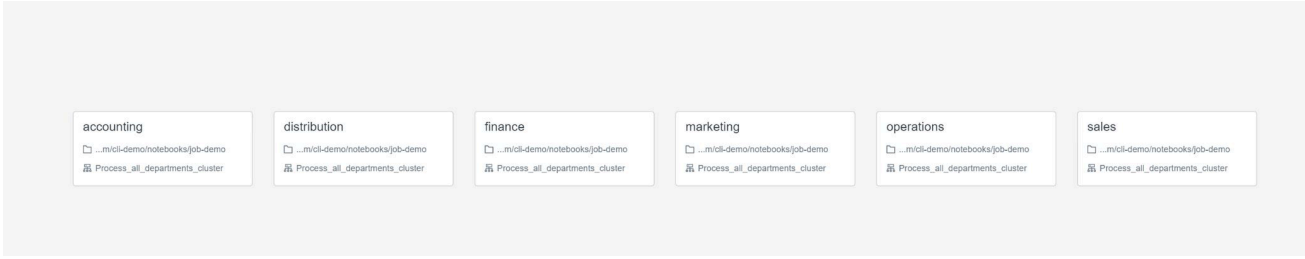
Unattempted

Here is how you setup

Create a single job and six tasks with the same notebook and assign a different parameter for each task ,



All tasks are added in a single job and can run parallel either using single shared cluster or with individual clusters.



38. Question

You are asked to setup two tasks in a databricks job, the first task runs a notebook to download the data from a remote system, and the second task is a DLT pipeline that can process this data, how do you plan to configure this in Jobs UI

- Single job cannot have a notebook task and DLT Pipeline task, use two different jobs with linear dependency.

Jobs UI does not support DTL pipeline, setup the first task using jobs UI and setup the DLT to run in continuous mode.

Jobs UI does not support DTL pipeline, setup the first task using jobs UI and setup the DLT to run in trigger mode.

Single job can be used to setup both notebook and DLT pipeline, use two different tasks with linear dependency.

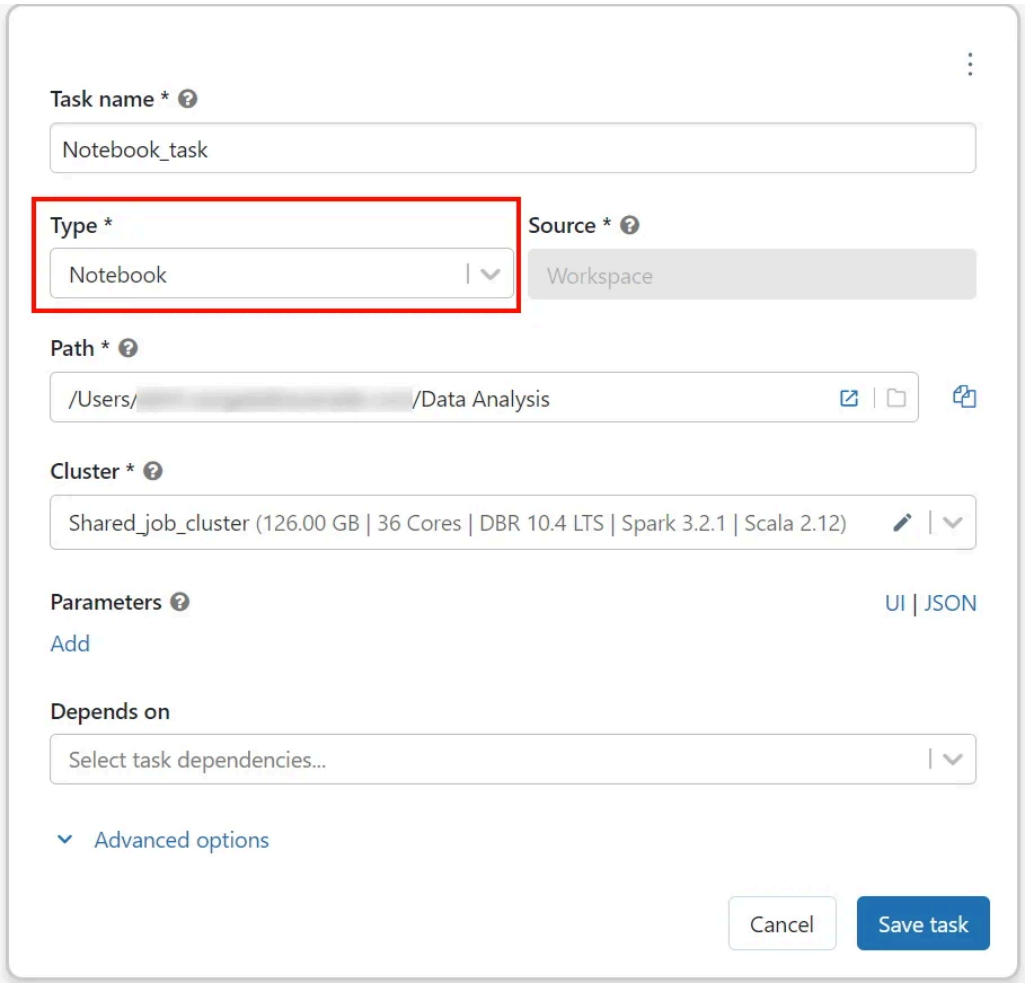
Add first step in the DLT pipeline and run the DLT pipeline as triggered mode in JOBS UI

Unattempted

The answer is Single job can be used to set up both notebook and DLT pipeline, use two different tasks with linear dependency,

Here is the JOB UI

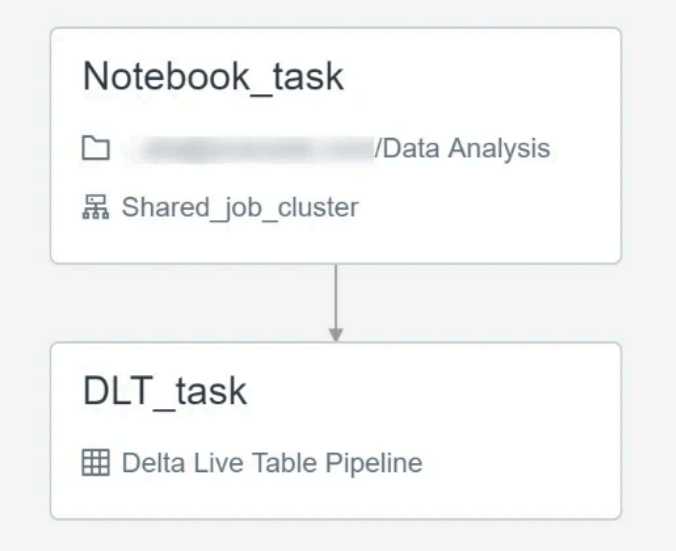
1. Create a notebook task



2. Create DLT task

a. add notebook task as dependency

3. Final view



Create the notebook task

DLT task

Final view

39. Question

You are asked to set up an alert to notify in an email every time a KPI indicater increases beyond a threshold value, team also asked you to include the actual value in the alert email notification.

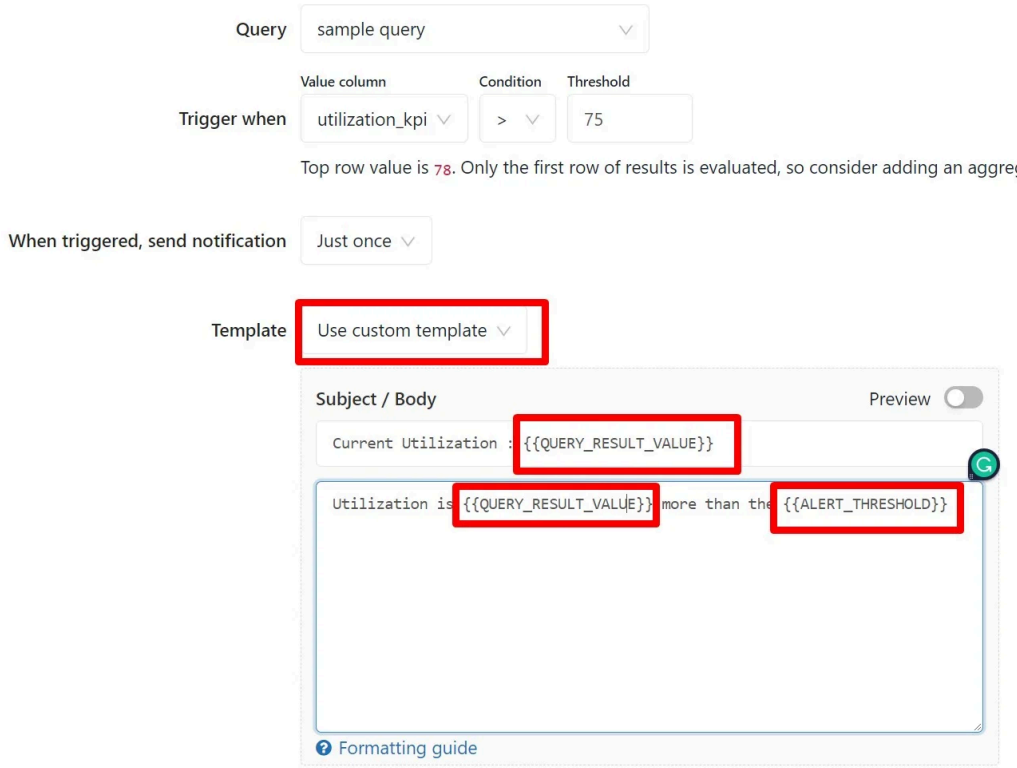
- Use notebook and python code to run every minute, using python variables to capture send the information in an email
- Setup an alert but use the default template to notify the message in email's subject
- **Setup an alert but use the custom template to notify the message in email's subject**
- Use the webhook destination instead so alert message can be customized
- Use custom email hook to customize the message

Unattempted

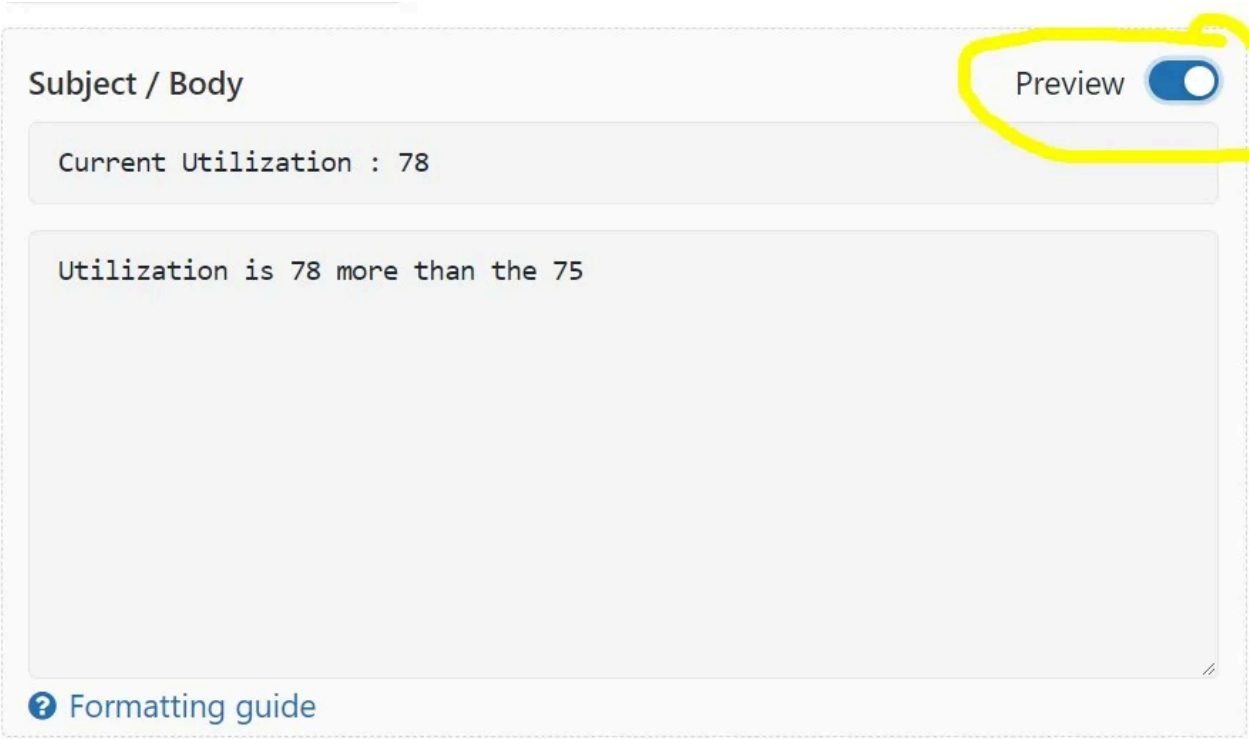
Alerts support custom template supports using variables to customize the default message, set up the query to compare the KPI current value to the threshold and use the variable QUERY_RESULT_VALUE to display the value in the email notification.

here is a simple alert, that uses variables in the custom template to present these values in the email notification message, when the alert is fired these variables get replaced with actual values.

Alert with custom template



When you enable preview you can see how the alert looks when you substitute the variables.



Below are additional template variables available to you with the custom template.

Alerts | Databricks on AWS

5. In the **Template** drop-down, choose a template:

- **Use default template**: Alert notification is a message with links to the Alert configuration screen and the Query screen.
- **Use custom template**: Alert notification includes more specific information about the alert.

  a. A box displays, consisting of input fields for subject and body. Any static content is valid, and you can incorporate built-in template variables:

  - `ALERT_STATUS`: The evaluated alert status (string).
  - `ALERT_CONDITION`: The alert condition operator (string).
  - `ALERT_THRESHOLD`: The alert threshold (string or number).
  - `ALERT_NAME`: The alert name (string).
  - `ALERT_URL`: The alert page URL (string).
  - `QUERY_NAME`: The associated query name (string).
  - `QUERY_URL`: The associated query page URL (string).
  - `QUERY_RESULT_VALUE`: The query result value (string or number).
  - `QUERY_RESULT_ROWS`: The query result rows (value array).
  - `QUERY_RESULT_COLS`: The query result columns (string array).

  An example subject, for instance, could be: `Alert "{{ALERT_NAME}}" changed status to {{ALERT_STATUS}}`.

  b. Click the **Preview** toggle button to preview the rendered result.

40. Question

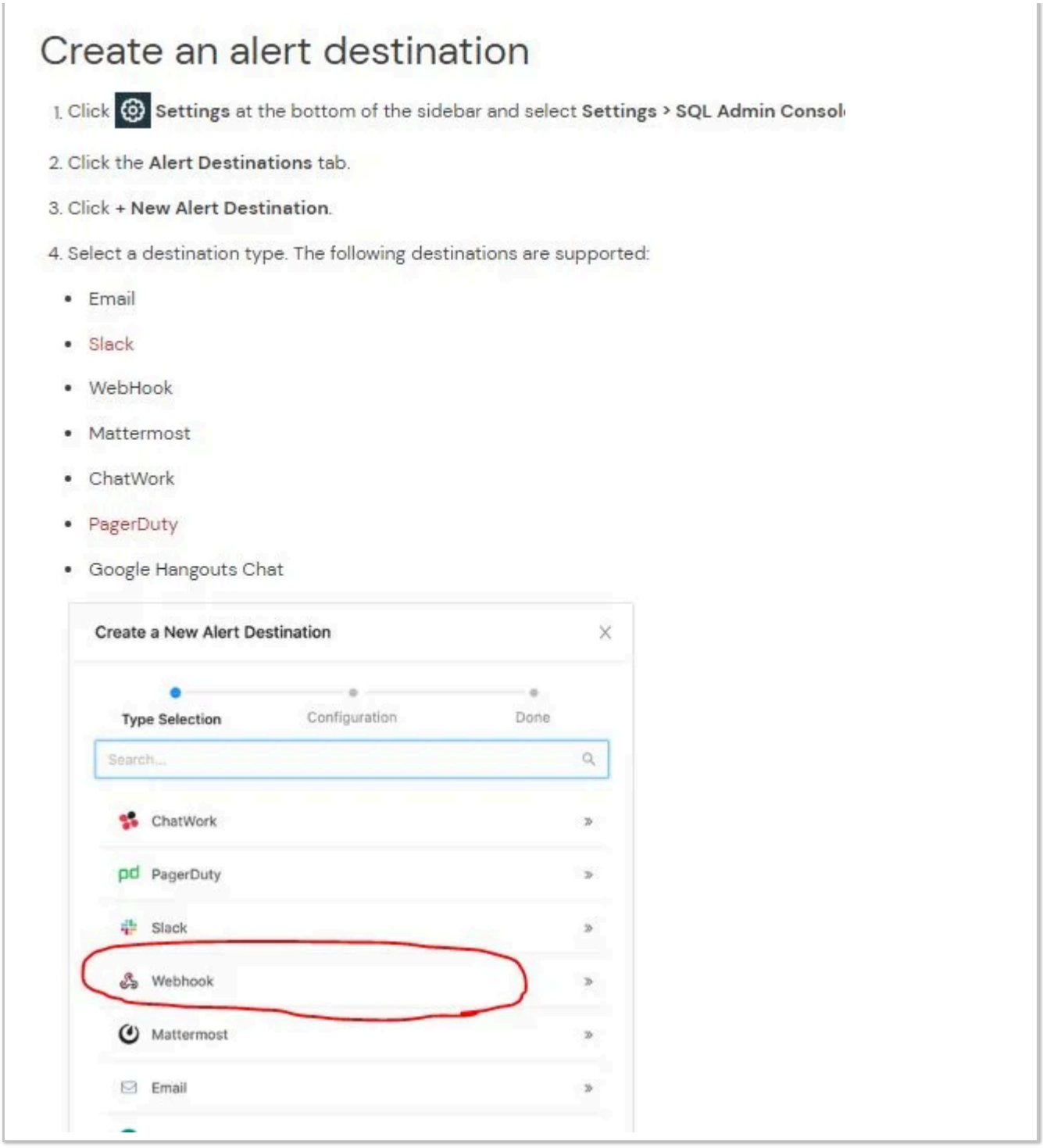
Operations team is using a centralized data quality monitoring system, a user can publish data quality metrics through a webhook, you were asked to develop a process to send messages using a webhook if there is atleast one duplicate record, which of the following approaches can be taken to integrate an alert with current data quality monitoring system

- Use notebook and Jobs to use python to publish DQ metrics
- Setup an alert to send an email, use python to parse email, and publish a webhook message
- Setup an alert with custom template
- **Setup an alert with custom Webhook destination**
- Setup an alert with dynamic template

Unattempted

Alerts supports multiple destinations, email is the default destination.

Alert destinations | Databricks on AWS



41. Question

You are currently working with the application team to setup a SQL Endpoint point, once the team started consuming the SQL Endpoint you noticed that during peak hours as the number of concurrent users increases you are seeing degradation in the query performance and the same queries are taking longer to run, which of the following steps can be taken to resolve the issue?

- They can turn on the Serverless feature for the SQL endpoint.
- **They can increase the maximum bound of the SQL endpoint's scaling range.**
- They can increase the cluster size(2X-Small to 4X-Large) of the SQL endpoint.
- They can turn on the Auto Stop feature for the SQL endpoint.

They can turn on the Serverless feature for the SQL endpoint and change the Spot Instance Policy from “Cost optimized” to “Reliability Optimized.”

Unattempted

The answer is, They can increase the maximum bound of the SQL endpoint's scaling range, when you increase the max scaling range more clusters are added so queries instead of waiting in the queue can start running using available clusters, see below for more explanation.

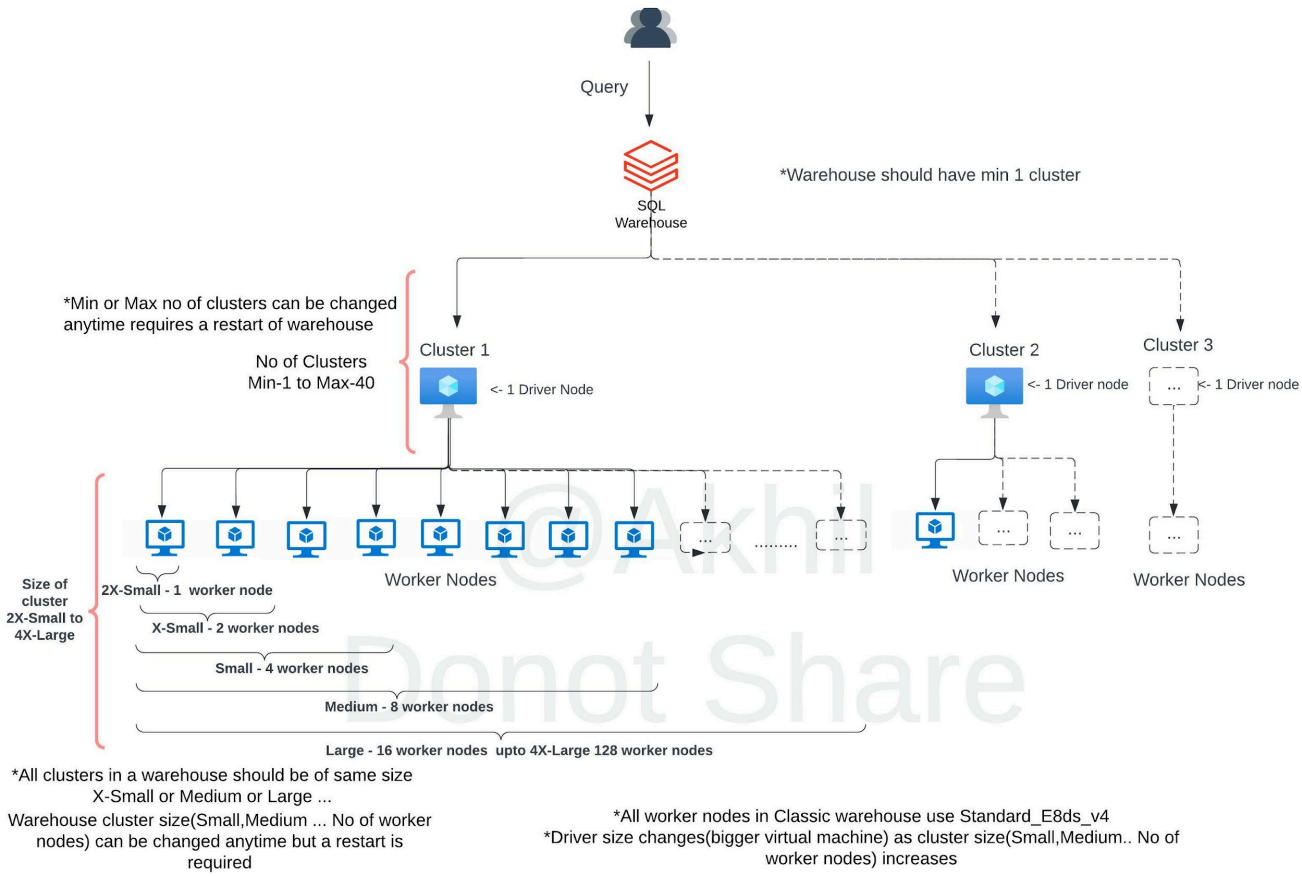
The question is looking to test your ability to know how to scale a SQL Endpoint(SQL Warehouse) and you have to look for cue words or need to understand if the queries are running sequentially or concurrently. if the queries are running sequentially then scale up(Size of the cluster from 2X-Small to 4X-Large) if the queries are running concurrently or with more users then scale out(add more clusters).

SQL Endpoint(SQL Warehouse) Overview: (Please read all of the below points and the below diagram to understand)

1. A SQL Warehouse should have at least one cluster
2. A cluster comprises one driver node and one or many worker nodes
3. No of worker nodes in a cluster is determined by the size of the cluster (2X -Small ->1 worker, X-Small ->2 workers___. up to 4X-Large -> 128 workers) this is called Scale up
4. A single cluster irrespective of cluster size(2X-Smal.. to ___4XLarge) can only run 10 queries at any given time if a user submits 20 queries all at once to a warehouse with 3X-Large cluster size and cluster scaling (min 1, max1) while 10 queries will start running the remaining 10 queries wait in a queue for these 10 to finish.
5. Increasing the Warehouse cluster size can improve the performance of a query, example if a query runs for 1 minute in a 2X-Small warehouse size, it may run in 30 Seconds if we change the warehouse size to X-Small. this is due to 2X-Small has 1 worker node and X-Small has 2 worker nodes so the query has more tasks and runs faster (note: this is an ideal case example, the scalability of a query performance depends on many factors, it can not always be linear)
6. A warehouse can have more than one cluster this is called Scale out. If a warehouse is configured with X-Small cluster size with cluster scaling(Min1, Max 2) Databricks spins up an additional cluster if it detects queries are waiting in the queue, If a warehouse is configured to run 2 clusters(Min1, Max 2), and let's say a user submits 20 queries, 10 queriers will start running and holds the remaining in the queue and databricks will automatically start the second cluster and starts redirecting the 10 queries waiting in the queue to the second cluster.
7. A single query will not span more than one cluster, once a query is submitted to a cluster it will remain in that cluster until the query execution finishes irrespective of how many clusters are available to scale.

Please review the below diagram to understand the above concepts:

SQL Warehouse Overview

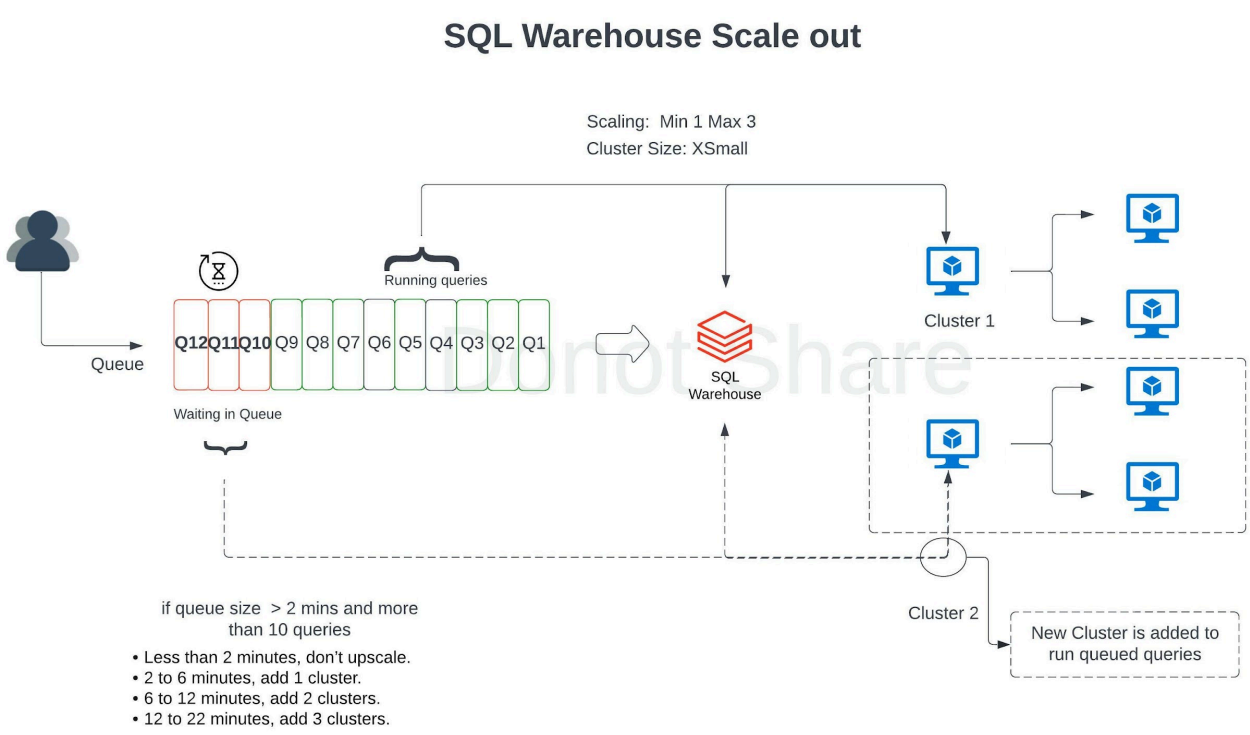


SQL endpoint(SQL Warehouse) scales horizontally(scale-out) and vertical (scale-up), you have to understand when to use what.

Scale-out -> to add more clusters for a SQL endpoint, change max number of clusters

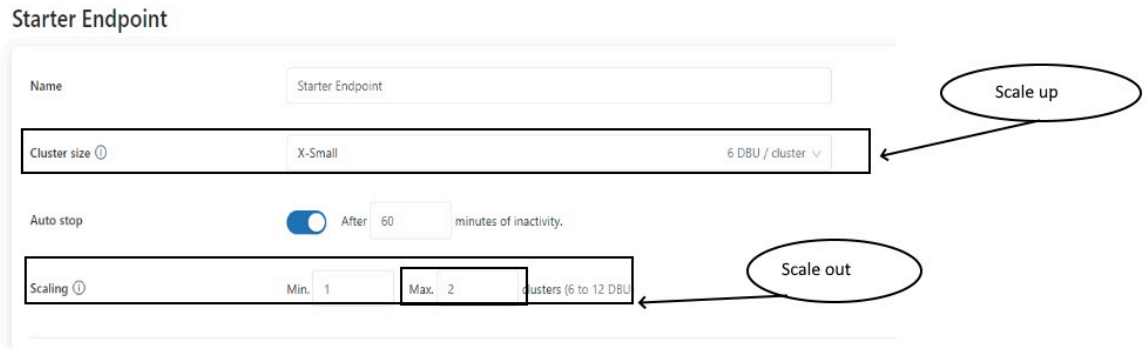
If you are trying to improve the throughput, being able to run as many queries as possible then having an additional cluster(s) will improve the performance.

Databricks SQL automatically scales as soon as it detects queries are in queuing state, in this example scaling is set for min 1 and max 3 which means the warehouse can add three clusters if it detects queries are waiting.



*** Please note 2 mins wait is only for classic warehouses for server less the wait can be much less.**

During the warehouse creation or after you have the ability to change the warehouse size (2X-Small...to ...4XLarge) to improve query performance and the maximize scaling range to add more clusters on a SQL Endpoint(SQL Warehouse) scale-out, if you are changing an existing warehouse you may have to restart the warehouse to make the changes effective.



42. Question

The data engineering team is using a bunch of SQL queries to review data quality and monitor the ETL job every day, which of the following approaches can be used to set up a schedule and automate this process?

- They can schedule the query to run every 1 day from the Jobs UI

They can schedule the query to refresh every 1 day from the query's page in Databricks SQL.

They can schedule the query to run every 12 hours from the Jobs UI.

They can schedule the query to refresh every 1 day from the SQL endpoint's page in Databricks SQL.

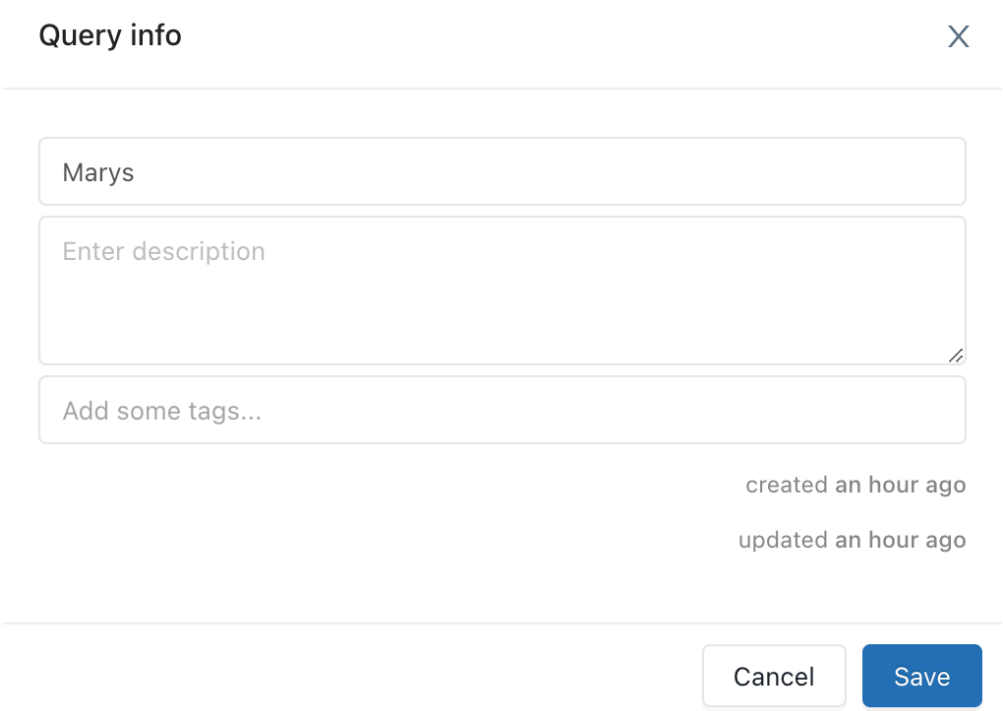
They can schedule the query to refresh every 12 hours from the SQL endpoint's page in Databricks SQL

Unattempted

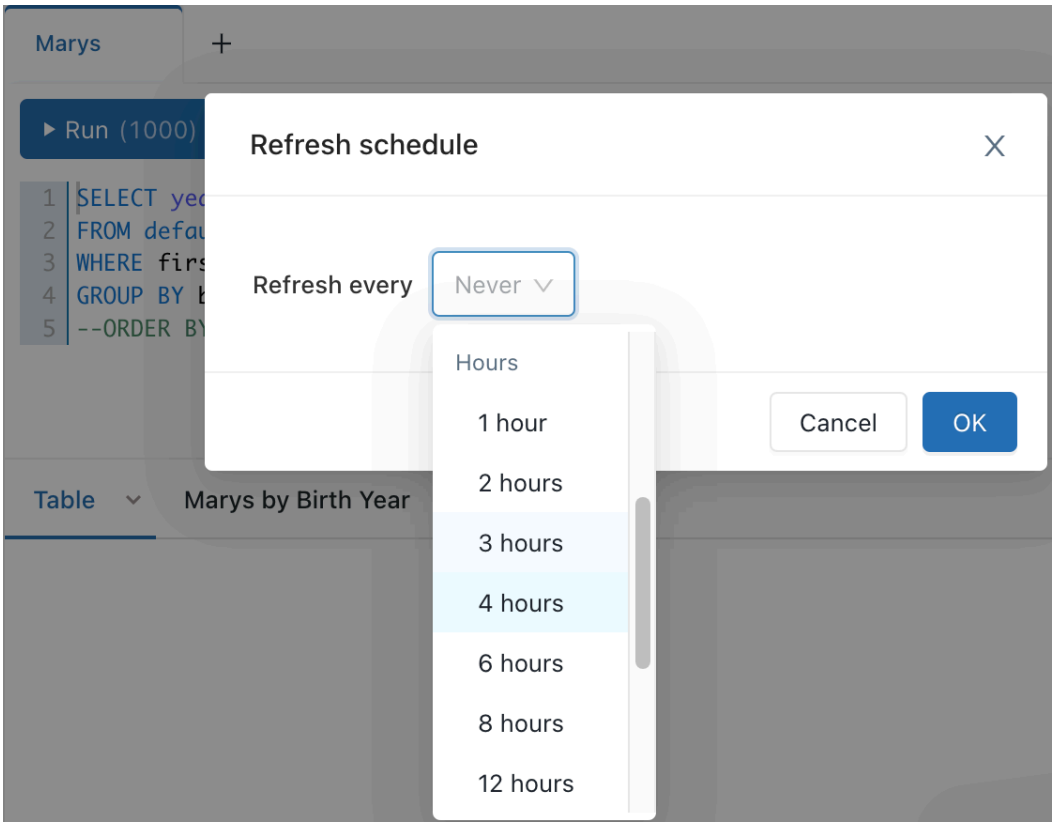
Individual queries can be refreshed on a schedule basis,

To set the schedule:

1. Click the query info tab.



2. Click the link to the right of Refresh Schedule to open a picker with schedule intervals.

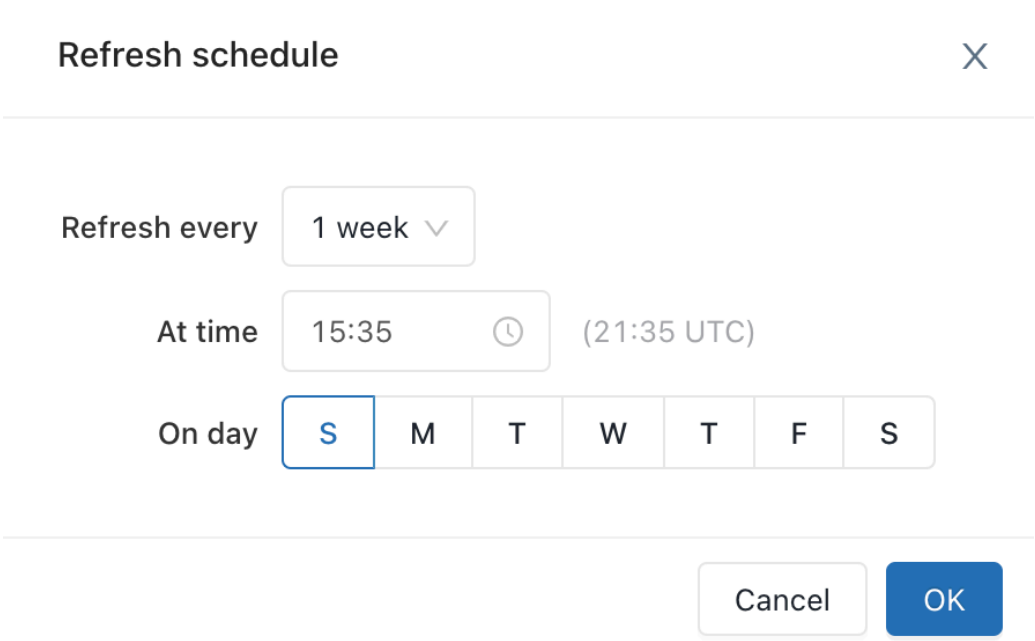


3. Set the schedule.

The picker scrolls and allows you to choose:

An interval: 1-30 minutes, 1-12 hours, 1 or 30 days, 1 or 2 weeks

A time. The time selector displays in the picker only when the interval is greater than 1 day and the day selection is greater than 1 week. When you schedule a specific time, Databricks SQL takes input in your computer's timezone and converts it to UTC. If you want a query to run at a certain time in UTC, you must adjust the picker by your local offset. For example, if you want a query to execute at 00:00 UTC each day, but your current timezone is PDT (UTC-7), you should select 17:00 in the picker:



4. Click OK.

Your query will run automatically.

If you experience a scheduled query not executing according to its schedule, you should manually trigger the query to make sure it doesn't fail. However, you should be aware of the following:

If you schedule an interval——for example, “every 15 minutes”——the interval is calculated from the last successful execution. If you manually execute a query, the scheduled query will not be executed until the interval has passed.
If you schedule a time, Databricks SQL waits for the results to be “outdated”. For example, if you have a query set to refresh every Thursday and you manually execute it on Wednesday, by Thursday the results will still be considered “valid”, so the query wouldn’t be scheduled for a new execution. Thus, for example, when setting a weekly schedule, check the last query execution time and expect the scheduled query to be executed on the selected day after that execution is a week old. Make sure not to manually execute the query during this time.
If a query execution fails, Databricks SQL retries with a back-off algorithm. The more failures the further away the next retry will be (and it might be beyond the refresh interval).
Refer documentation for additional info,
https://docs.microsoft.com/en-us/azure/databricks/sql/user/queries/schedule-query

43. Question

In order to use Unity catalog features, which of the following steps needs to be taken on managed/external tables in the Databricks workspace?

- Enable unity catalog feature in workspace settings
- **Migrate/upgrade objects in workspace managed/external tables/view to unity catalog**
- Upgrade to DBR version 15.0
- Copy data from workspace to unity catalog
- Upgrade workspace to Unity catalog

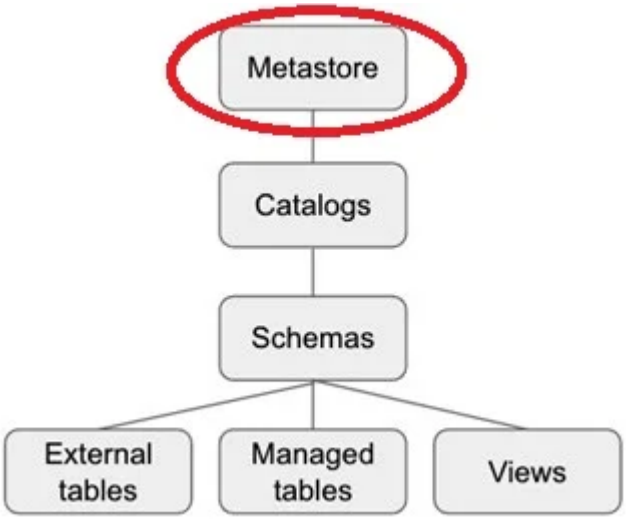
Unattempted
Upgrade tables and views to Unity Catalog – Azure Databricks | Microsoft Docs
Managed table: Upgrade a managed to Unity Catalog
External table: Upgrade an external table to Unity Catalog

44. Question

What is the top-level object in unity catalog?

- Catalog
- Table
- Workspace
- Database
- **Metastore**

Unattempted
Key concepts – Azure Databricks | Microsoft Docs



45. Question

One of the team members Steve who has the ability to create views, created a new view called regional_sales_vw on the existing table called sales which is owned by John, and the second team member Kevin who works with regional sales managers wanted to query the data in regional_sales_vw, so Steve granted the permission to Kevin using command
GRANT VIEW, USAGE ON regional_sales_vw to kevin@company.com but Kevin is still unable to access the view?

- Kevin needs select access on the table sales
- Kevin needs owner access on the view regional_sales_vw
- **Steve is not the owner of the sales table**
- Kevin is not the owner of the sales table
- Table access control is not enabled on the table and view

Unattempted
Ownership determines whether or not you can grant privileges on derived objects to other users, since Steve is not the owner of the underlying sales table, he can not grant access to the table or data in the table indirectly.
Only owner(user or group) can grant access to a object
https://docs.microsoft.com/en-us/azure/databricks/security/access-control/table-acls/object-privileges#a-user-has-select-privileges-on-a-view-of-table-t-but-when-that-user-tries-to-select-from-that-view-they-get-the-error-user-does-not-have-privilege-select-on-table
Data object privileges – Azure Databricks | Microsoft Doc

46. Question

Kevin is the owner of the schema sales, Steve wanted to create new table in sales schema called regional_sales so Kevin grants the create table permissions to Steve. Steve creates the new table called regional_sales in sales schema, who is the owner of the table regional_sales

- Kevin is the owner of sales schema, all the tables in the schema will be owned by Kevin

- Steve is the owner of the table
- By default ownership is assigned DBO
- By default ownership is assigned to DEFAULT_OWNER
- Kevin and Smith both are owners of table

Unattempted

A user who creates the object becomes its owner, does not matter who is the owner of the parent object.

47. Question

You were asked to setup a new all-purpose cluster, but the cluster is unable to start which of the following steps do you need to take to identify the root cause of the issue and the reason why the cluster was unable to start?

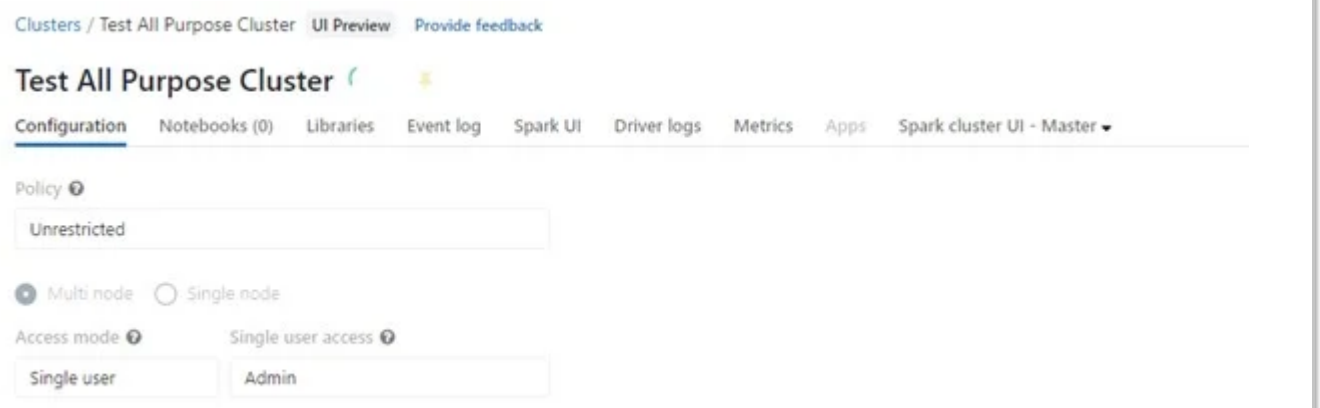
- Check the cluster driver logs
- Check the cluster event logs
- Workspace logs
- Storage account
- Data plane

Unattempted

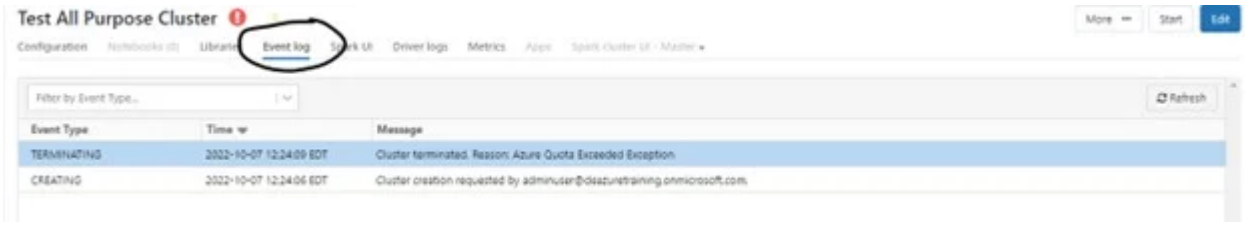
Cluster event logs are very useful, to identify issues pertaining to cluster availability. Cluster may not start due to resource limitations or issues with the cloud providers.

Some of the common issues include a subnet for compute VM reaching its limits or exceeding the subscription or cloud account CPU quota limit.

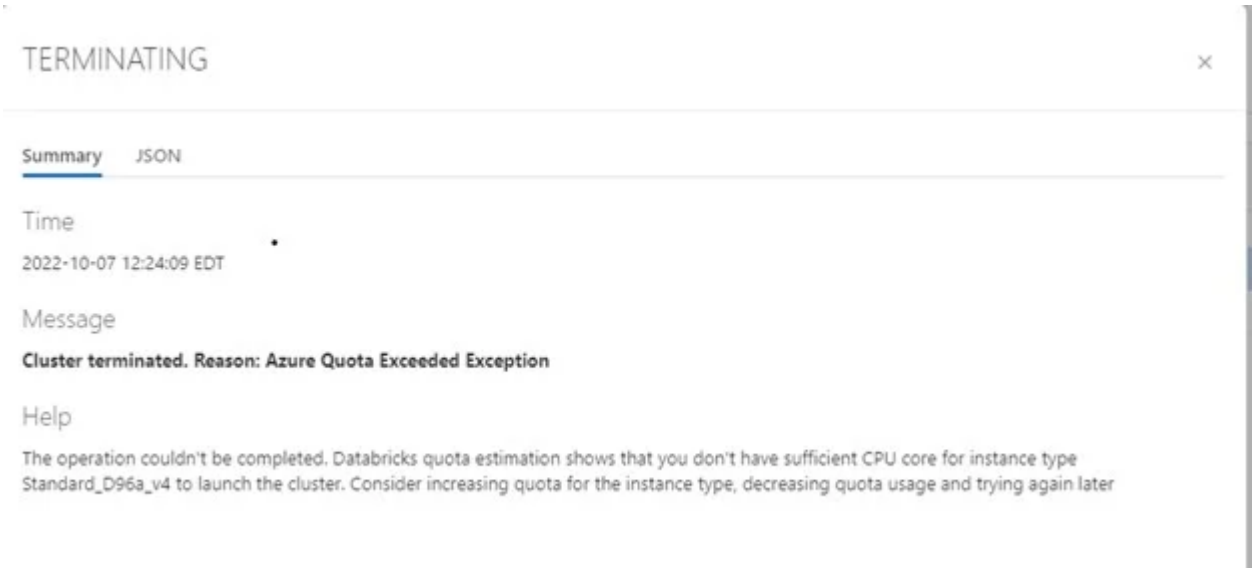
Here is an example where the cluster did not start due to subscription reaching the quota limit on a certain type of cpu cores for a VM type.



Click on event logs



Click on the message to see the detailed error message on why the cluster did not start.



48. Question

Which of the following developer operations in CI/CD flow can be implemented in Databricks Repos?

- Delete branch
- Trigger Databricks CICD pipeline
- Commit and push code
- Create a pull request
- Approve the pull request

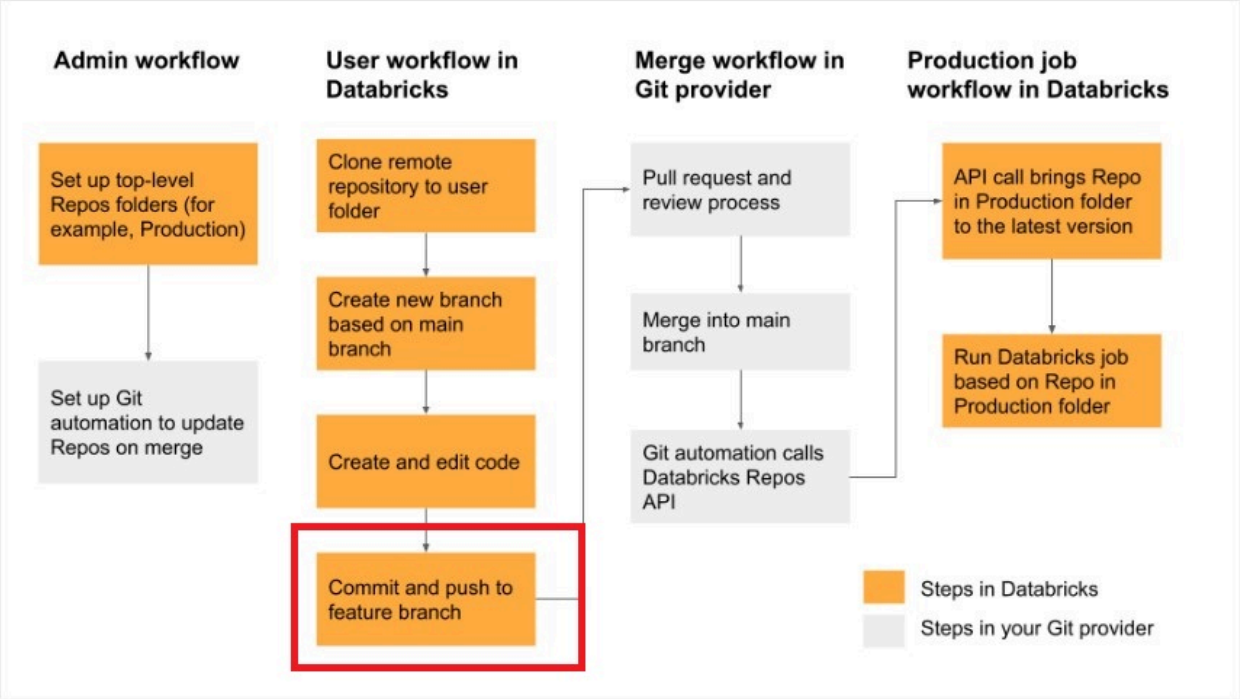
Unattempted

The answer is Commit and push code.

See the below diagram to understand the role Databricks Repos and Git provider plays when building a CI/CD workflow.

All the steps highlighted in yellow can be done Databricks Repo, all the steps highlighted in Gray are done in a git provider like Github or Azure Devops.

Exam focus: Please study the below image carefully to understand all of the steps in the CI/CD flow to understand the tasks that are implemented in Databricks Repo vs Git Provider, exam may ask a different type of questions based on this flow.



49. Question

You noticed that a team member started using an all-purpose cluster to develop a notebook and used the same all-purpose cluster to set up a job that can run every 30 mins so they can update underlying tables which are used in a dashboard. What would you recommend for reducing the overall cost of this approach?

- Reduce the size of the cluster
- Reduce the number of nodes and enable auto scale
- Enable auto termination after 30 mins
- **Change the cluster all-purpose to job cluster when scheduling the job**
- Change the cluster mode from all-purpose to single-mode

Unattempted

While using an all-purpose cluster is ok during development but anytime you don't need to interact with a notebook, especially for a scheduled job it is less expensive to use a job cluster. Using an all-purpose cluster can be twice as expensive as a job cluster.

Please note: The compute cost you pay the cloud provider for the same cluster type and size between an all-purpose cluster and job cluster is the same the only difference is the DBU cost.

The total cost of cluster = Total cost of VM compute(Azure or AWS or GCP) + Cost per DBU

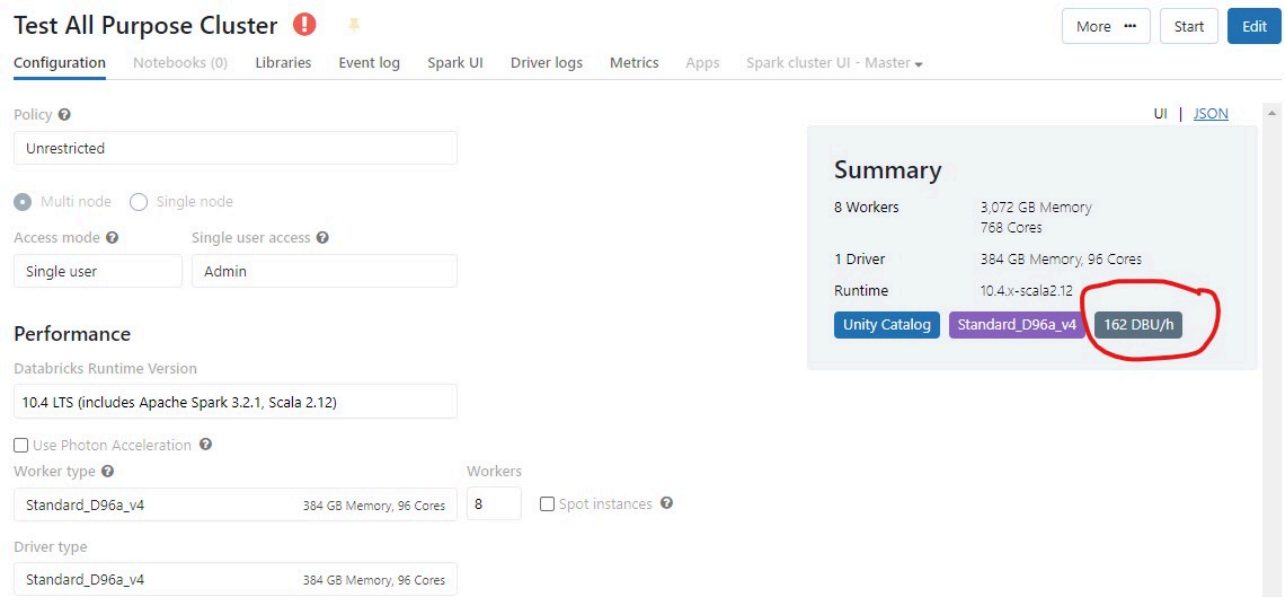
The per DBU cost varies between all-purpose and Job Cluster

Here is the recent cost estimate from AWS between Jobs Cluster and all-purpose Cluster, for jobs compute its $0.15 cents per DBU v$0.55 cents per DBU for all-purpose

aws	Standard	Premium	Enterprise
	One platform for your data analytics and ML workloads	Data analytics and ML at scale across your business	Data analytics and ML for your mission critical workloads
CLASSIC COMPUTE — Jobs Light Compute Run data engineering pipelines to build data lakes.	$0.07 / DBU	$0.10 / DBU	$0.13 / DBU
Jobs Compute Jobs Compute Photon Run data engineering pipelines to build data lakes and manage data at scale.	$0.10 / DBU	$0.15 / DBU	$0.20 / DBU
Delta Live Tables Delta Live Tables Photon Easily build high quality streaming or batch ETL pipelines using Python or SQL with the DLT Edition that is best for your workload. Learn more	$0.20 - $0.36 / DBU	$0.20 - $0.36 / DBU	$0.20 - $0.36 / DBU
SQL Compute Run SQL queries for BI reporting, analytics and visualization to get timely insights from data lakes.	-	$0.22 / DBU	$0.22 / DBU
All-Purpose Compute All-Purpose Compute Photon Run interactive data science and machine learning workloads. Also good for data engineering, BI and data analytics.	$0.40 / DBU	$0.55 / DBU	$0.65 / DBU

How do I check how much the DBU cost for my cluster?

When you click on an exister cluster or when you look at the cluster details you will see this in the top right corner



50. Question

Which of the following commands can be used to run one notebook from another notebook?

- notebook.utils.run("full notebook path")

- execute.utils.run("full notebook path")
- **dbutils.notebook.run("full notebook path")**
- only job clusters can run notebook
- spark.notebook.run("full notebook path")

Unattempted

The answer is dbutils.notebook.run(" full notebook path ")

Here is the full command with additional options.

run(path: String, timeout_seconds: int, arguments: Map): String

dbutils.notebook.run("ful-notebook-name", 60, {"argument": "data", "argument2": "data2", ...})

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